MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

COMPARATOR, MAGNITUDE, DIGITAL, KEY CODE JHB

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic magnitude comparator module.

1.2 Classification This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to. Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3 3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.0749 failures per million operating hours for class I or 0.1760 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3 4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be JHB.

3.5.2 Size. The module size shall be 1A

3 5 3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3 5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A

3 5 6 Depth The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/327-1 for class I and M28787/327-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3 6 Circuit pin configuration The function diagram and pin assignments shall conform to figure 1.

3 7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3 8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2)

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5 5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply

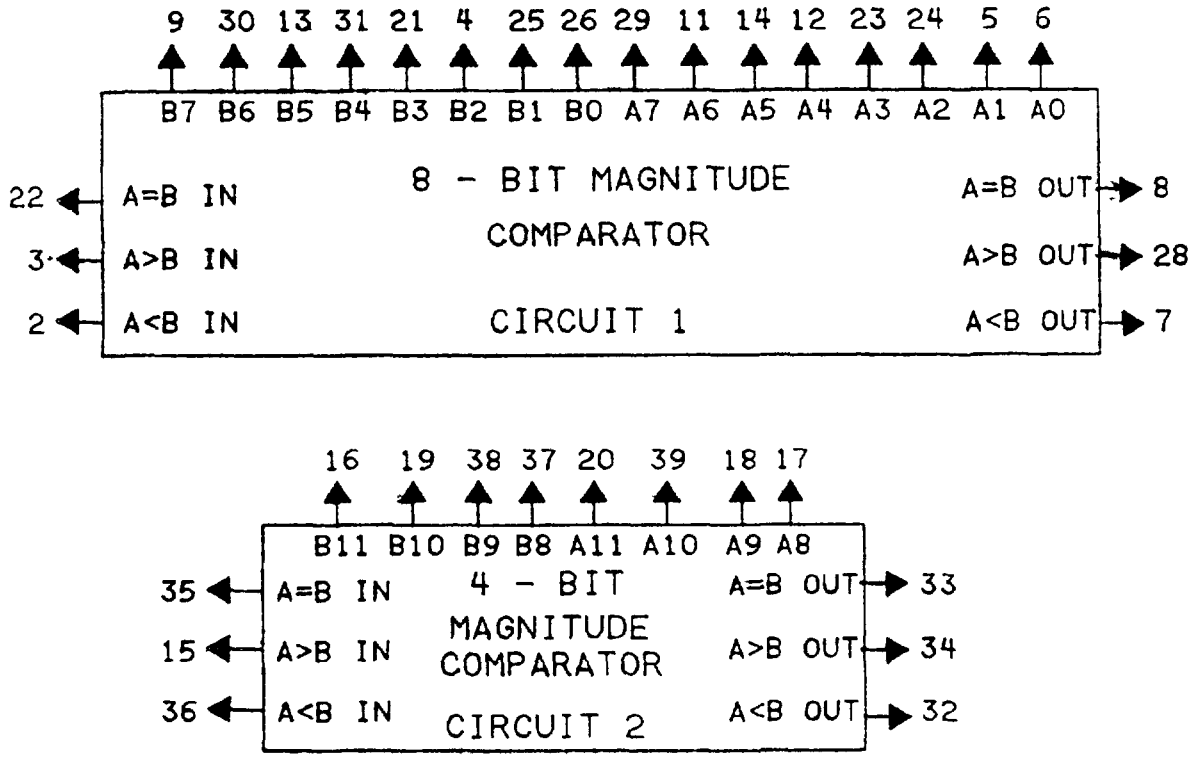
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4 4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787

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A0 - A11, B0 - B11 - COMPARING INPUTS
A = B IN, A>B IN, A<B IN - CASCADING INPUTS
A = B OUT, A>B OUT, A<B OUT - OUTPUTS

FIGURE 1. JHB module function diagram, magnitude comparator.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1, A2	Isolation													
	Initial	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B	V_{CC} stress	IV	S	← Module must pass remaining tests →										
C1-C30	Input breakdown current	V	S	0	1	0	1	0	1	0	1	0	1	mA
D1-D6	Output short circuit current	VI	X	-40	-100	-40	-100	-40	-100	-40	-100	-40	-100	mA
E1-E30	Input clamp diode voltage	VII	X	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	V
F1-F26	Input current, high level	VIII	X	0	150	0	150	0	150	0	150	0	150	µA
F27-F30	Input current, high level	VIII	X	0	50	0	50	0	50	0	50	0	50	µA
G1-G26	Input current, low level	VIII	X	-0.1	-6.6	-0.1	-6.6	-0.1	-6.6	-0.1	-6.6	-0.1	-6.6	mA
G27-G30	Input current, low level	VIII	X	-0.1	-2 2	-0.1	-2.2	-0.1	-2.2	-0.2	-2.2	-0.2	-2.2	mA
H1-H79	Functional compare	IX	X	← See table IX →										
J1-J6	Output voltage, high level	X	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
K1-K6	Output voltage, low level	X	X	0	500	0	500	0	500	0	500	0	500	mV
L	Power supply current	XI	S	30	345	30	345	30	345	30	345	30	345	mA
M1-M4,M11-M14, M19-M22,M27-M30,M35-M38, M43-M46,M59-M62,M75-M78, M91-M94	Propagation delay time (t_{PLH}), C_L = 50 pF	XII	S	2	19	2	19	2	21	2	19	2	23	ns
M5	Propagation delay time (t_{PLH}), C_L = 50 pF	XII	S	2	26	2	26	2	29	2	26	2	32	ns
M6-M9	Propagation delay time (t_{PLH}), C_L = 50 pF	XII	S	2.0	9.5	2 0	9.5	2.0	10.5	2.0	9.5	2.0	11.5	ns

See footnotes at end of table.

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TABLE I Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
M10	Propagation delay time (t_{PLH}), C_L = 50 pF	XII	S	2	13	2	13	2.0	14.5	2	13	2	16	ns
M15-M18,M23-M26,M31-M34,M39-M42,M51-M54,M67-M70,M83-M86,M99-M102	Propagation delay time (t_{PLH}), C_L = 50 pF	XII	S	2	21	2	21	2	23	2	21	2	25	ns
M47-M50,M63-M66,M79-M82,M95-M98	Propagation delay time (t_{PLH}), C_L = 50 pF	XII	S	2 0	28.5	2.0	28.5	2.0	31.5	2.0	28.5	2.0	34.5	ns
M55-M58,M71-M74,M87-M90,M103-M106	Propagation delay time (t_{PLH}), C_L = 50 pF	XII	S	2	34	2	34	2.0	3.75	2	34	2	41	ns
N1-N4	Propagation delay time (t_{PHL}), C_L = 50 pF	XII	S	2	21	2	21	2.0	23.5	2	21	2	26	ns
N5	Propagation delay time (t_{PHL}), C_L = 50 pF	XII	S	2	19	2	19	2.0	21.5	2	19	2	24	ns
N6-N9	Propagation delay time (t_{PHL}), C_L = 50 pF	XII	S	2.0	10.5	2.0	10.5	2	12	2.0	10.5	2	13	ns
N10	Propagation delay time (t_{PHL}), C_L = 50 pF	XII	S	2.0	9.5	2.0	9.5	2	11	2.0	9.5	2	12	ns
N11-N46,N51-N54,N59-N62,N67-N70,N75-N78,N83-N86,N91-N94,N99-N102	Propagation delay time (t_{PHL}), C_L = 50 pF	XII	S	2.0	19.5	2.0	19.5	2	22	2.0	19.5	2	24	ns
N47-N50,N63-N66,N79-N82,N95-N98	Propagation delay time (t_{PHL}), C_L = 50 pF	XII	S	2	30	2	30	2	34	2	30	2	37	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
N55-N58,N71-N74,N87-N90, N103-N106	Propagation delay time (t_{PHL}), C_L = 50 pF	XII	S	2	29	2	29	2	33	2	29	2	36	ns
P	Power supply filter capacitance	XIII	S	1.6	3.0	1 6	3.0	1 6	3.0	1.6	3.0	1.6	3.0	µF
Q1-Q6	Output crosstalk, high level	XIV	Q	0	400	0	400	0	400	0	400	0	400	mV
R1-R6	Output crosstalk, low level	XIV	Q	0	500	0	500	0	500	0	500	0	500	mV
S	Power supply transient amplitude	XV	Q	0	500	0	500	0	500	0	500	0	500	mV
T1-T4	Input capacitance	XVI	Q	2	15	2	15	2	15	2	15	2	15	pF
T5-T30	Input capacitance	XVI	Q	2	20	2	20	2	20	2	20	2	20	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. If test conditions A, B, C, or D are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through S shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition P, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through S shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through T shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5 PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

6.4 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life limits from table I.

	Class I	Class II	Unit
Maximum power dissipation (outputs not loaded)	1.9	1 9	W
Input current (A, B & A=B)			
Low level (V_{IL} = 0.0 V)	-0.1 min, -6.6 max	-0.1 min, -6.6 max	mA
Low level (V_{IL} = 0.5 V)	-0.1 min, -6.0 max	-0.1 min, -6.0 max	mA

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	Class I	Class II	Unit
Input current			
High level (V_{IH} = 2.4 V)	0 min, 150 max	0 min, 150 max	µA
High level (V_{IH} = 5.5 V)	0 min, 1 max	0 min, 1 max	mA
Output drive current			
Low level (V_{OL} = 0.5 V)	20	20	mA
High level (V_{OH} = 2.5 V)	-1	-1	mA
Input current (A>B & A<B)			
Low level (V_{IL} = 0.0 V)	-0.1 min, -2.2 max	-0.1 min, -2.2 max	mA
(V_{IL} = 0.5 V)	-0.1 min, -2.0 max	-0.1 min, -2.0 max	mA
High level (V_{IH} = 2.4 V)	0 min, 50 max	0 min, 50 max	µA
(V_{IH} = 5.5 V)	0 min, 1 max	0 min, 1 max	mA

6.5 Input to output relationship. Each magnitude comparator module has the input to output relationship of table VII for positive logic.

6 5.1 Parallel outputs. Paralleling of outputs is not permissible.

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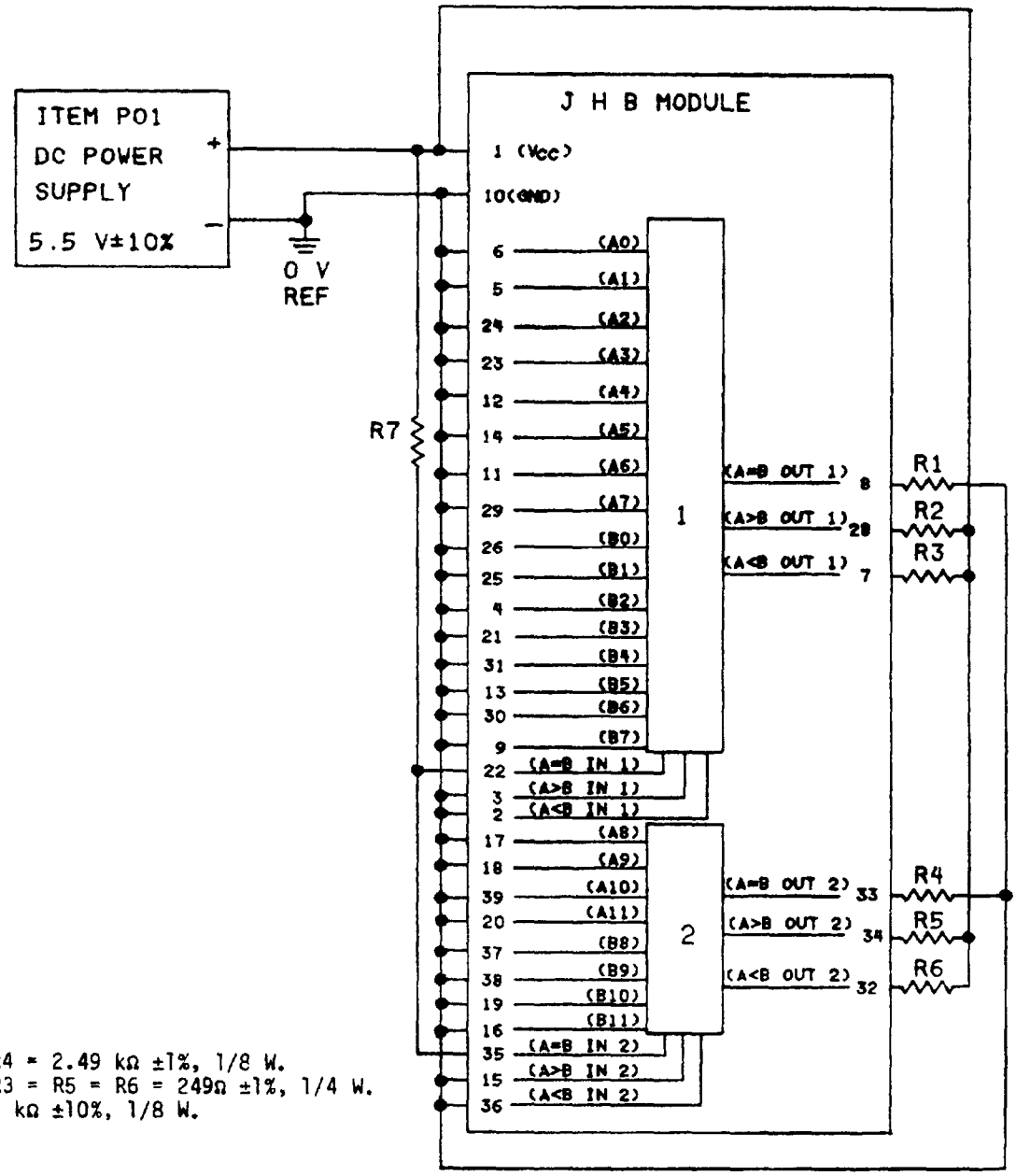
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TABLE II. Input to output relationship. 1/

Comparing inputs												Cascading inputs			Outputs		
A11,B11	A10,B10	A9,B9	A8,B8	A7,B7	A6,B6	A5,B5	A4,B4	A3,B3	A2,B2	A1,B1	A0,B0	A>B	A<B	A=B	A>B	A<B	A=B
A11>B11	X	X	X									X	X	X	H	L	L
A11<B11	X	X	X									X	X	X	L	H	L
A11=B11	A10>B10	X	X									X	X	X	H	L	L
A11=B11	A10<B10	X	X									X	X	X	L	H	L
A11=B11	A10=B10	A9>B9	X									X	X	X	H	L	L
A11=B11	A10=B10	A9<B9	X									X	X	X	L	H	L
A11=B11	A10=B10	A9=B9	A8>B8									X	X	X	H	L	L
A11=B11	A10=B10	A9=B9	A8<B8									X	X	X	L	H	L
A11=B11	A10=B10	A9=B9	A8=B8									H	L	L	H	L	L
A11=B11	A10=B10	A9=B9	A8=B8									L	H	L	L	H	L
A11=B11	A10=B10	A9=B9	A8=B8									L	L	H	L	L	H
				A7>B7	X	X	X	X	X	X	X	X	X	X	H	L	L
				A7<B7	X	X	X	X	X	X	X	X	X	X	L	H	L
				A7=B7	A6>B6	X	X	X	X	X	X	X	X	X	H	L	L
				A7=B7	A6<B6	X	X	X	X	X	X	X	X	X	L	H	L
				A7=B7	A6=B6	A5>B5	X	X	X	X	X	X	X	X	H	L	L
				A7=B7	A6=B6	A5<B5	X	X	X	X	X	X	X	X	L	H	L
				A7=B7	A6=B6	A5=B5	A4>B4	X	X	X	X	X	X	X	H	L	L
				A7=B7	A6=B6	A5=B5	A4<B4	X	X	X	X	X	X	X	L	H	L
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				A7=B7	A6=B6	A5=B5	A4=B4	A3<B3	X	X	X	X	X	X	L	H	L
				A7=B7	A6=B6	A5=B5	A4=B4	A3=B3	A2>B2	X	X	X	X	X	H	L	L
				A7=B7	A6=B6	A5=B5	A4=B4	A3=B3	A2<B2	X	X	X	X	X	L	H	L
				A7=B7	A6=B6	A5=B5	A4=B4	A3=B3	A2=B2	A1>B1	X	X	X	X	H	L	L
				A7=B7	A6=B6	A5=B5	A4=B4	A3=B3	A2=B2	A1<B1	X	X	X	X	L	H	L
				A7=B7	A6=B6	A5=B5	A4=B4	A3=B3	A2=B2	A1=B1	A0>B0	X	X	X	H	L	L
				A7=B7	A6=B6	A5=B5	A4=B4	A3=B3	A2=B2	A1=B1	A0<B0	X	X	X	L	H	L
				A7=B7	A6=B6	A5=B5	A4=B4	A3=B3	A2=B2	A1=B1	A0=B0	H	L	L	H	L	L
				A7=B7	A6=B6	A5=B5	A4=B4	A3=B3	A2=B2	A1=B1	A0=B0	L	H	L	L	H	L
				A7=B7	A6=B6	A5=B5	A4=B4	A3=B3	A2=B2	A1=B1	A0=B0	L	L	H	L	L	H

1/ H = high level.
L = low level.
X = don't care.

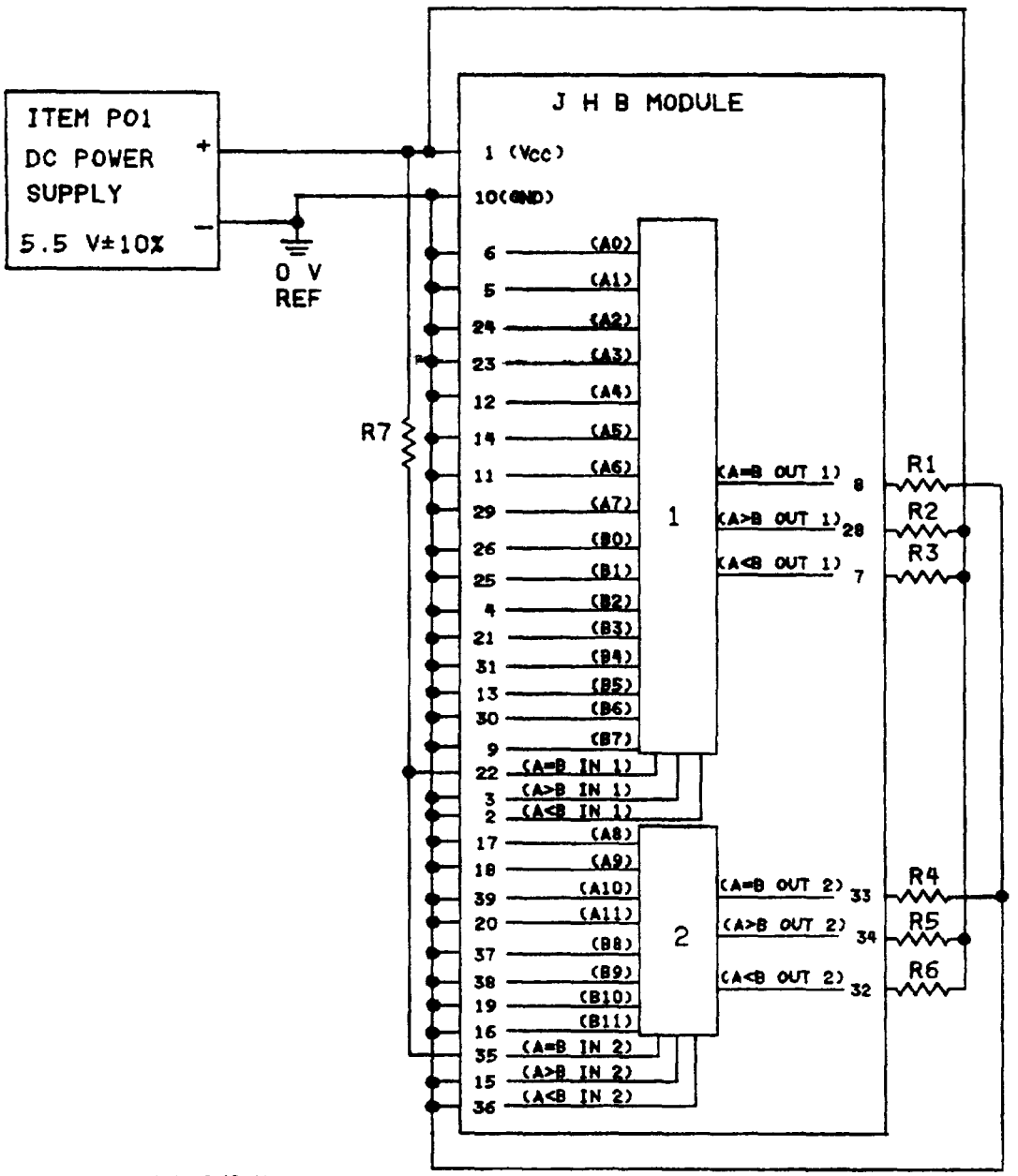
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NOTES.
1. R1 = R4 = 2.49 kΩ ±1%, 1/8 W.
2. R2 = R3 = R5 = R6 = 249Ω ±1%, 1/4 W.
3. R7 = 1 kΩ ±10%, 1/8 W.

FIGURE 2. Operating temperature and life test circuit.

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NOTES:
1. R1 = R4 = 2.49 kΩ ±1%, 1/8 W.
2. R2 = R3 = R5 = R6 = 249Ω ±1%, 1/4 W.
3. R7 = 1 kΩ ±10%, 1/8 W.

FIGURE 3. Component temperature test circuit.

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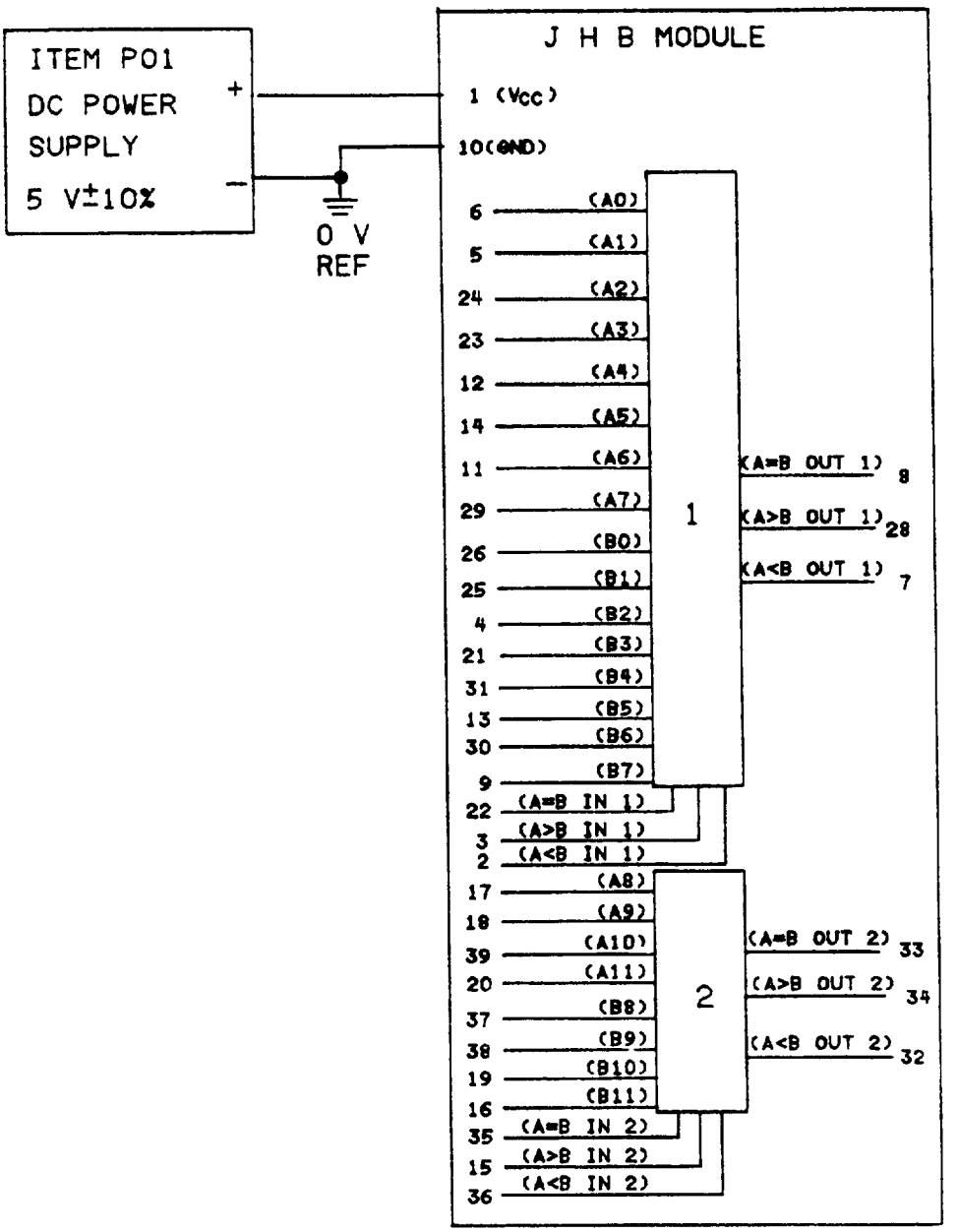


FIGURE 4. Vibration test circuit.

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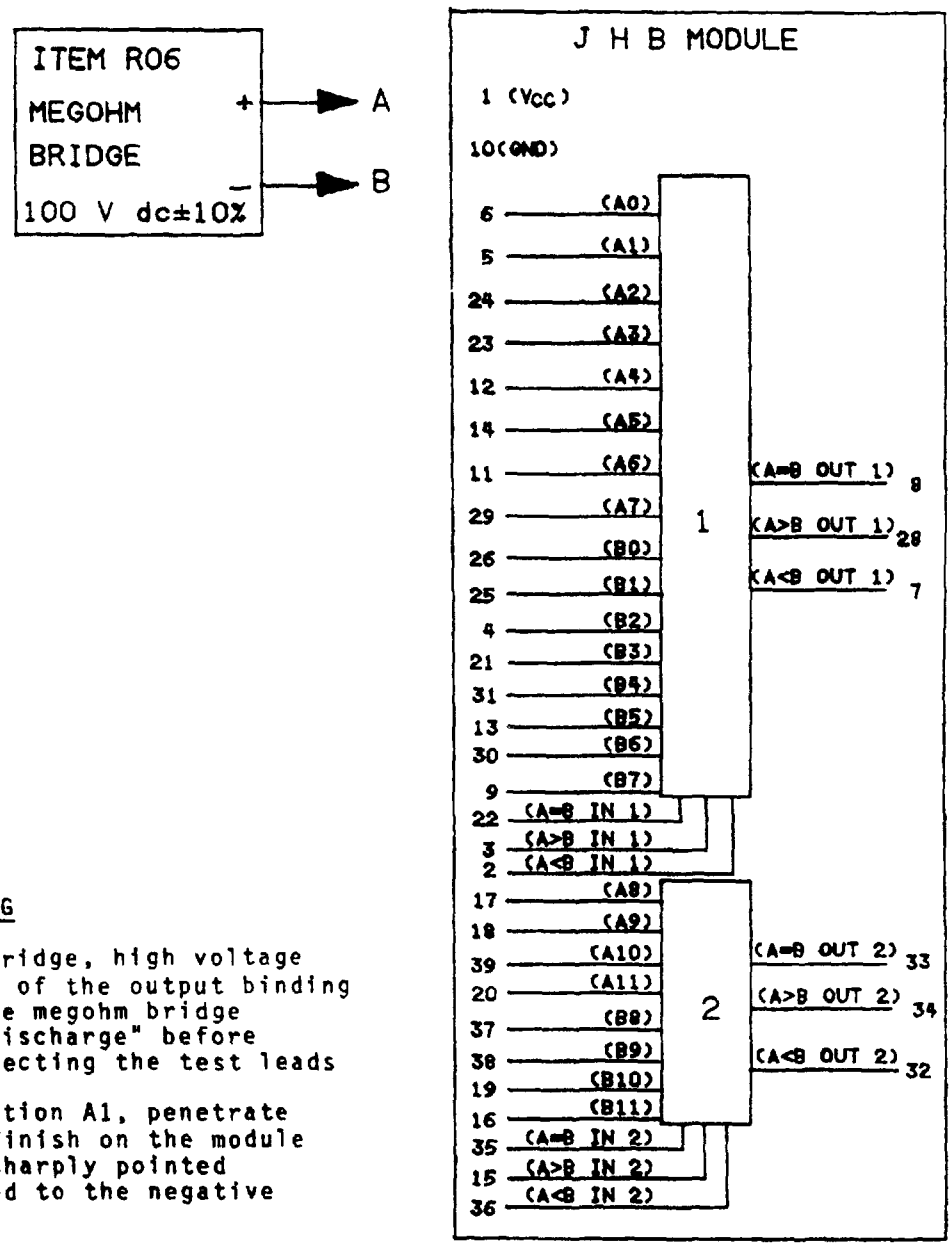
TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation	All pins	Frame	10	∞	MΩ	5
A2	Isolation	All pins except 27,40	27,40	10	∞	MΩ	5

TABLE IV. V_{CC} stress test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
B	V_{CC} stress	Must pass remaining tests			6

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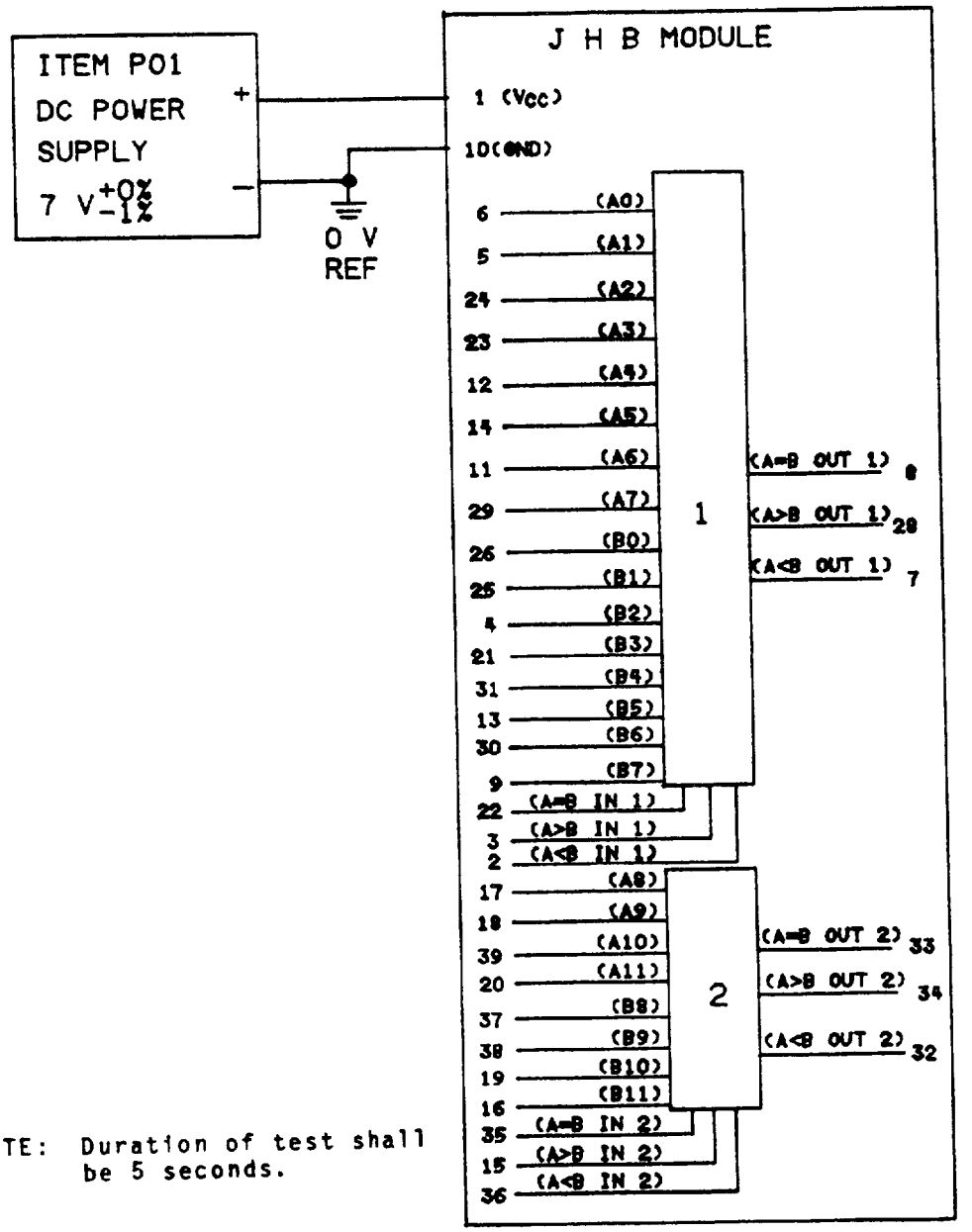
WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads

NOTE: For test condition A1, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 5. Isolation test circuit for table III.

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NOTE: Duration of test shall be 5 seconds.

FIGURE 6. V_{CC} stress test circuit for table IV.

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TABLE V Input breakdown current test.

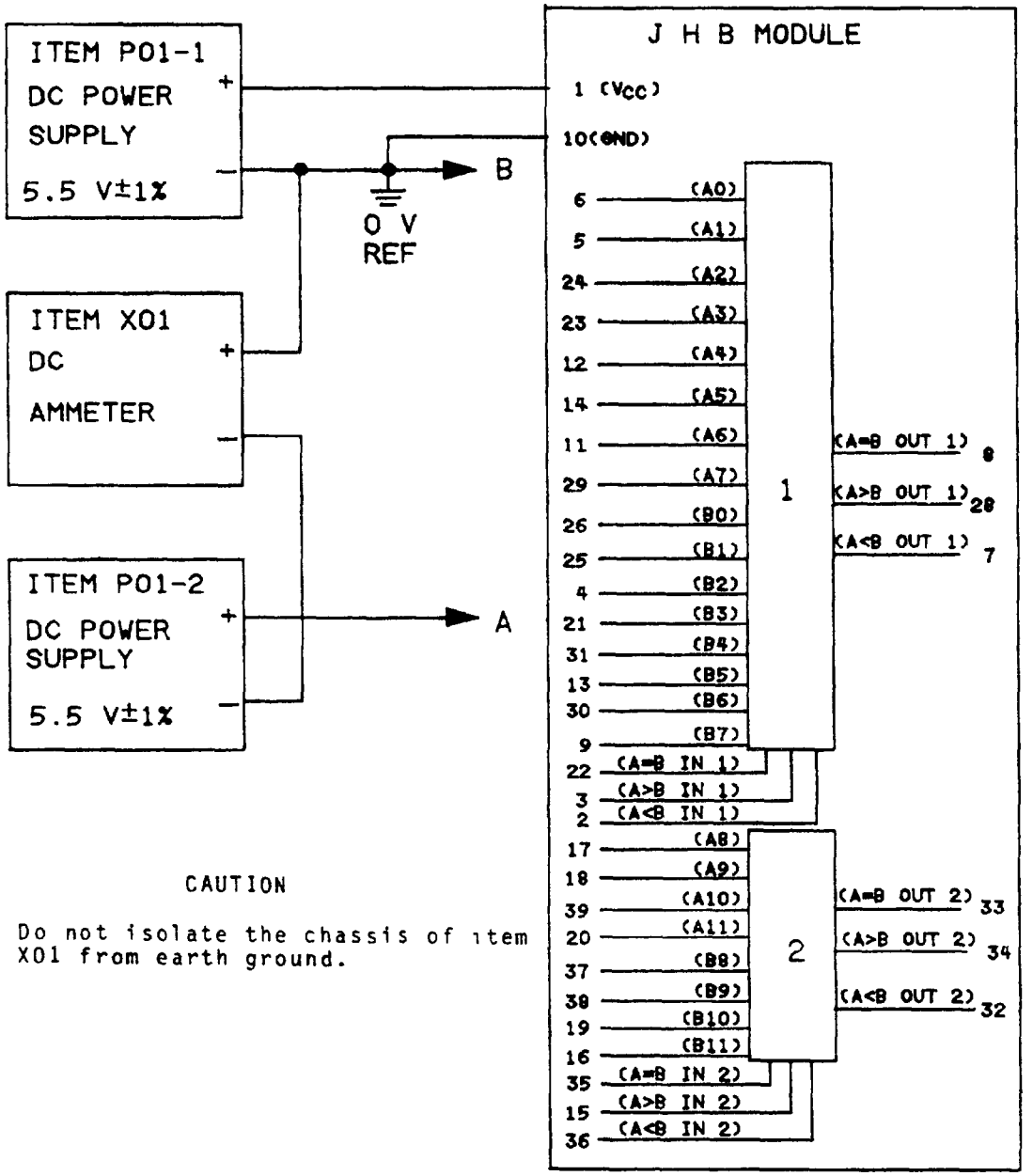
Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C1	A0	Input breakdown current	6	26	0	1	mA	7
C2	A1	"	5	25	"	"	"	"
C3	A2	"	24	4	"	"	"	"
C4	A3	"	23	21	"	"	"	"
C5	A4	"	12	31	"	"	"	"
C6	A5	"	14	13	"	"	"	"
C7	A6	"	11	30	"	"	"	"
C8	A7	"	29	9	"	"	"	"
C9	A8	"	17	37	"	"	"	"
C10	A9	"	18	38	"	"	"	"
C11	A10	"	39	19	"	"	"	"
C12	A11	"	20	16	"	"	"	"
C13	B0	"	26	6	"	"	"	"
C14	B1	"	25	5	"	"	"	"
C15	B2	"	4	24	"	"	"	"
C16	B3	"	21	23	"	"	"	"
C17	B4	"	31	12	"	"	"	"
C18	B5	"	13	14	"	"	"	"
C19	B6	"	30	11	"	"	"	"
C20	B7	"	9	29	"	"	"	"
C21	B8	"	37	17	"	"	"	"

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TABLE V. Input breakdown current test - Continued.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C22	B9	Input breakdown current	38	18	0	1	mA	7
C23	B10	"	19	39	"	"	"	"
C24	B11	"	16	20	"	"	"	"
C25	A=B IN 1	"	22	26	"	"	"	"
C26	A=B IN 2	"	35	37	"	"	"	"
C27	A<B IN 1	"	2	26	"	"	"	"
C28	A>B IN 1	"	3	26	"	"	"	"
C29	A>B IN 2	"	15	37	"	"	"	"
C30	A<B IN 2	"	36	37	"	"	"	"

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NOTE. Item PO1-2 shall be current limited to 5 mA.

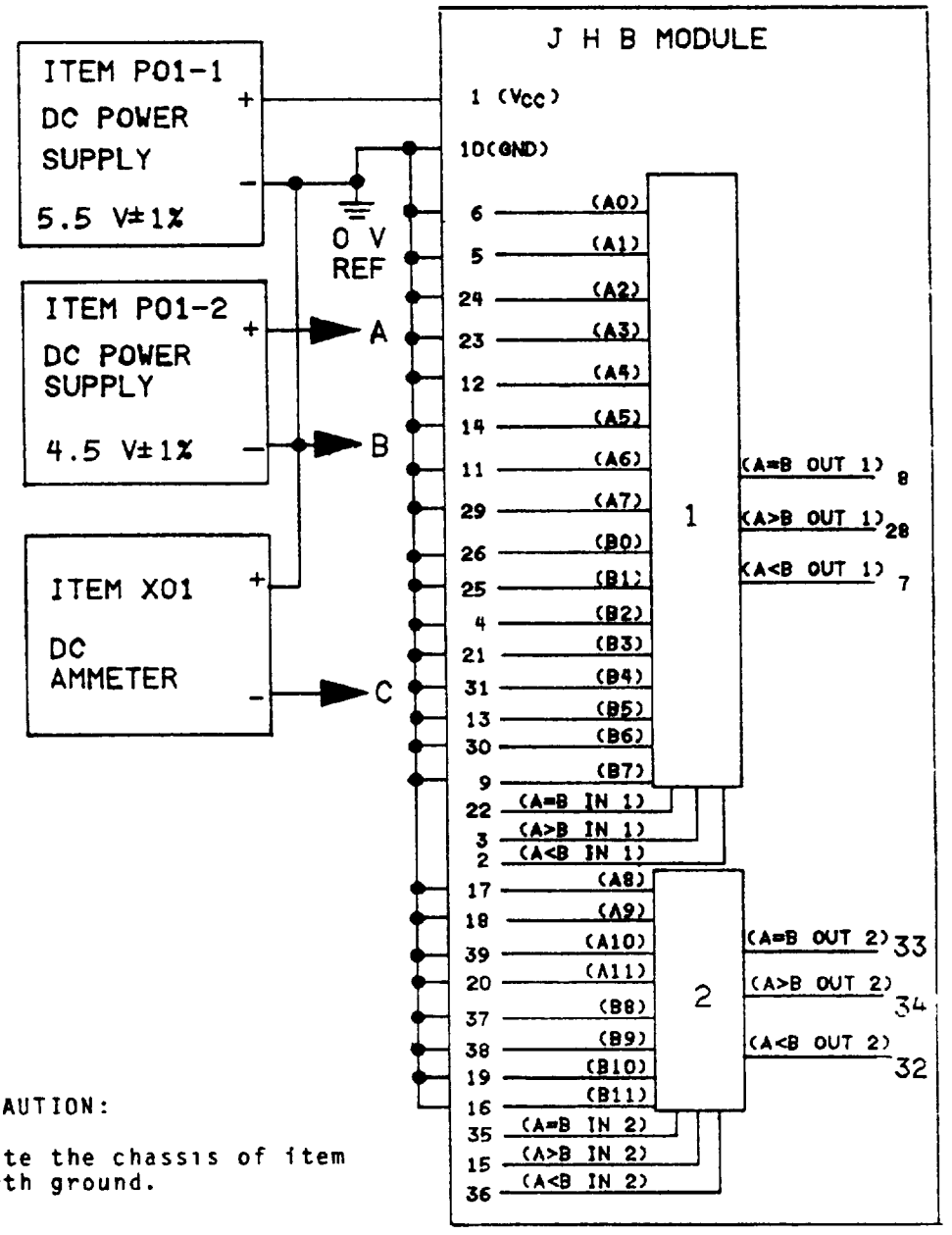
FIGURE 7. Input breakdown current test circuit for table V.

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TABLE VI. Output short circuit current test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
D1	A=B OUT 1 Output short circuit current	22	2,3	8	-40	-100	mA	8
D2	A>B OUT 1 "	3	2,22	28	"	"	"	"
D3	A<B OUT 1 "	2	3,22	7	"	"	"	"
D4	A=B OUT 2 "	35	15,36	33	"	"	"	"
D5	A>B OUT 2 "	15	35,36	34	"	"	"	"
D6	A<B OUT 2 "	36	15,35	32	"	"	"	"

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CAUTION:

Do not isolate the chassis of item X01 from earth ground.

NOTE. For output short circuit, only one output shall be shorted at a time.

FIGURE 8. Output short circuit current test circuit for table VI.

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TABLE VII. Input clamp diode voltage test.

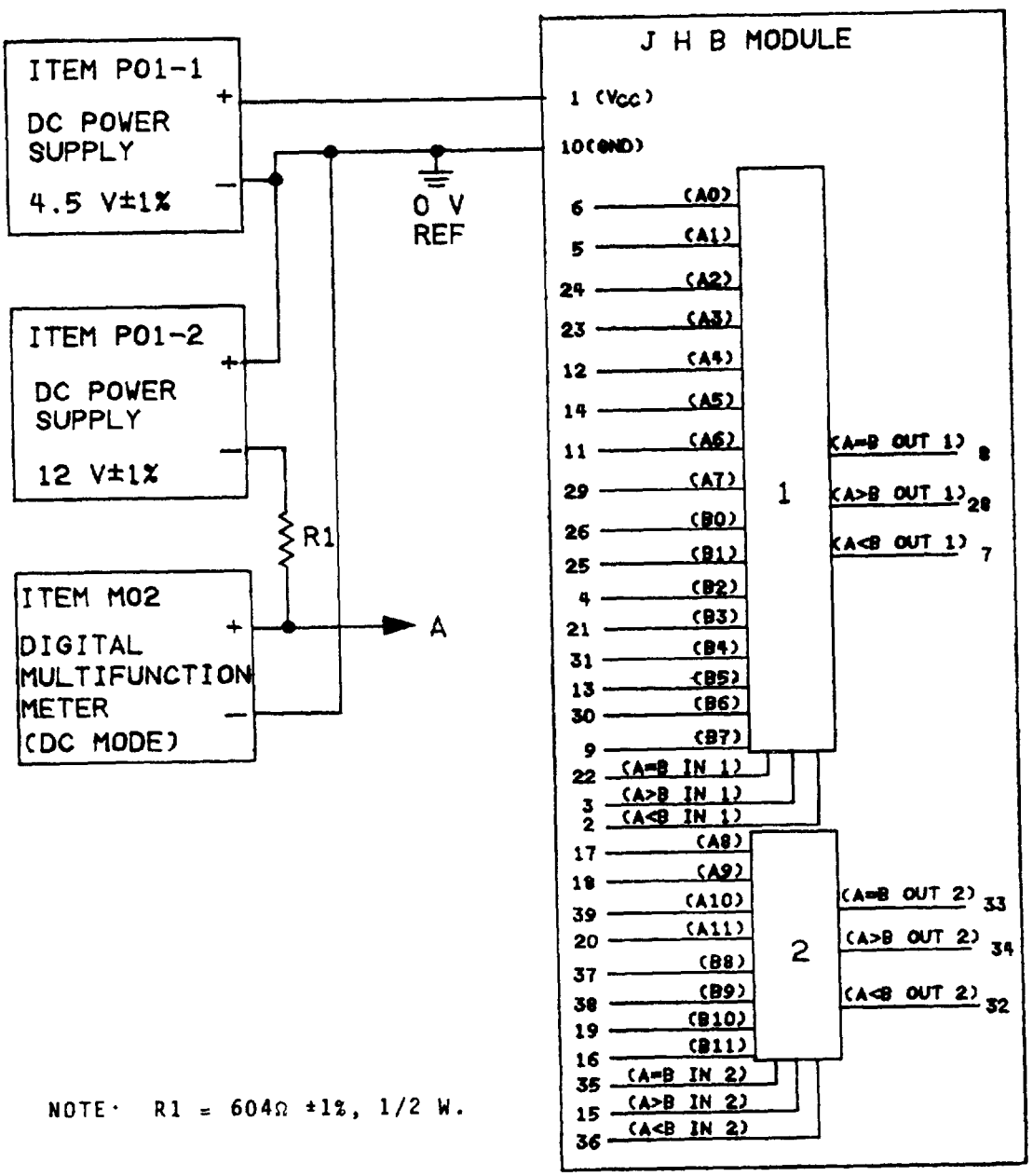
Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
E1	A0 Input clamp diode voltage	6	-0.4	-1.2	V	9
E2	A1 "	5	"	"	"	"
E3	A2 "	24	"	"	"	"
E4	A3 "	23	"	"	"	"
E5	A4 "	12	"	"	"	"
E6	A5 "	14	"	"	"	"
E7	A6 "	11	"	"	"	"
E8	A7 "	29	"	"	"	"
E9	A8 "	17	"	"	"	"
E10	A9 "	18	"	"	"	"
E11	A10 "	39	"	"	"	"
E12	A11 "	20	"	"	"	"
E13	B0 "	26	"	"	"	"
E14	B1 "	25	"	"	"	"
E15	B2 "	4	"	"	"	"
E16	B3 "	21	"	"	"	"
E17	B4 "	31	"	"	"	"
E18	B5 "	13	"	"	"	"
E19	B6 "	30	"	"	"	"
E20	B7 "	9	"	"	"	"
E21	B8 "	37	"	"	"	"

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TABLE VII. Input clamp diode voltage test - Continued.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
E22	B9 Input clamp diode voltage	38	-0.4	-1.2	V	9
E23	B10 "	19	"	"	"	"
E24	B11 "	16	"	"	"	"
E25	A=B IN 1 "	22	"	"	"	"
E26	A>B IN 1 "	3	"	"	"	"
E27	A<B IN 1 "	2	"	"	"	"
E28	A=B IN 2 "	35	"	"	"	"
E29	A>B IN 2 "	15	"	"	"	"
E30	A<B IN 2 "	36	"	"	"	"

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NOTE: R1 = 604Ω ±1%, 1/2 W.

FIGURE 9. Input clamp diode voltage test circuit for table VII.

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TABLE VIII. Input current (high-level and low-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F1	A0 Input current, high level	6	Open	1/	0	150	µA	10
F2	A1 "	5	"	"	"	"	"	"
F3	A2 "	24	"	"	"	"	"	"
F4	A3 "	23	"	"	"	"	"	"
F5	A4 "	12	"	"	"	"	"	"
F6	A5 "	14	"	"	"	"	"	"
F7	A6 "	11	"	"	"	"	"	"
F8	A7 "	29	"	"	"	"	"	"
F9	A8 "	17	"	"	"	"	"	"
F10	A9 "	18	"	"	"	"	"	"
F11	A10 "	39	"	"	"	"	"	"
F12	A11 "	20	"	"	"	"	"	"
F13	B0 "	26	"	"	"	"	"	"
F14	B1 "	25	"	"	"	"	"	"
F15	B2 "	4	"	"	"	"	"	"
F16	B3 "	21	"	"	"	"	"	"
F17	B4 "	31	"	"	"	"	"	"
F18	B5 "	13	"	"	"	"	"	"
F19	B6 "	30	"	"	"	"	"	"
F20	B7 "	9	"	"	"	"	"	"
F21	B8 "	37	"	"	"	"	"	"

See footnote at end of table

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TABLE VIII. Input current (high-level and low-level) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
F22	B9	Input current, high level	38	Open	1/	0	150	µA	10
F23	B10	"	19	"	"	"	"	"	"
F24	B11	"	16	"	"	"	"	"	"
F25	A=B IN 1	"	22	"	"	"	"	"	"
F26	A=B IN 2	"	35	"	"	"	"	"	"
F27	A>B IN 1	"	3	"	"	"	50	"	"
F28	A<B IN 1	"	2	"	"	"	"	"	"
F29	A>B IN 2	"	15	"	"	"	"	"	"
F30	A<B IN 2	"	36	"	"	"	"	"	"
G1	A0	Input current, low level	Open	6	Open	-0.1	-6.6	mA	"
G2	A1	"	"	5	"	"	"	"	"
G3	A2	"	"	24	"	"	"	"	"
G4	A3	"	"	23	"	"	"	"	"
G5	A4	"	"	12	"	"	"	"	"
G6	A5	"	"	14	"	"	"	"	"
G7	A6	"	"	11	"	"	"	"	"
G8	A7	"	"	29	"	"	"	"	"
G9	A8	"	"	17	"	"	"	"	"
G10	A9	"	"	18	"	"	"	"	"
G11	A10	"	"	39	"	"	"	"	"
G12	A11	"	"	20	"	"	"	"	"

See footnote at end of table.

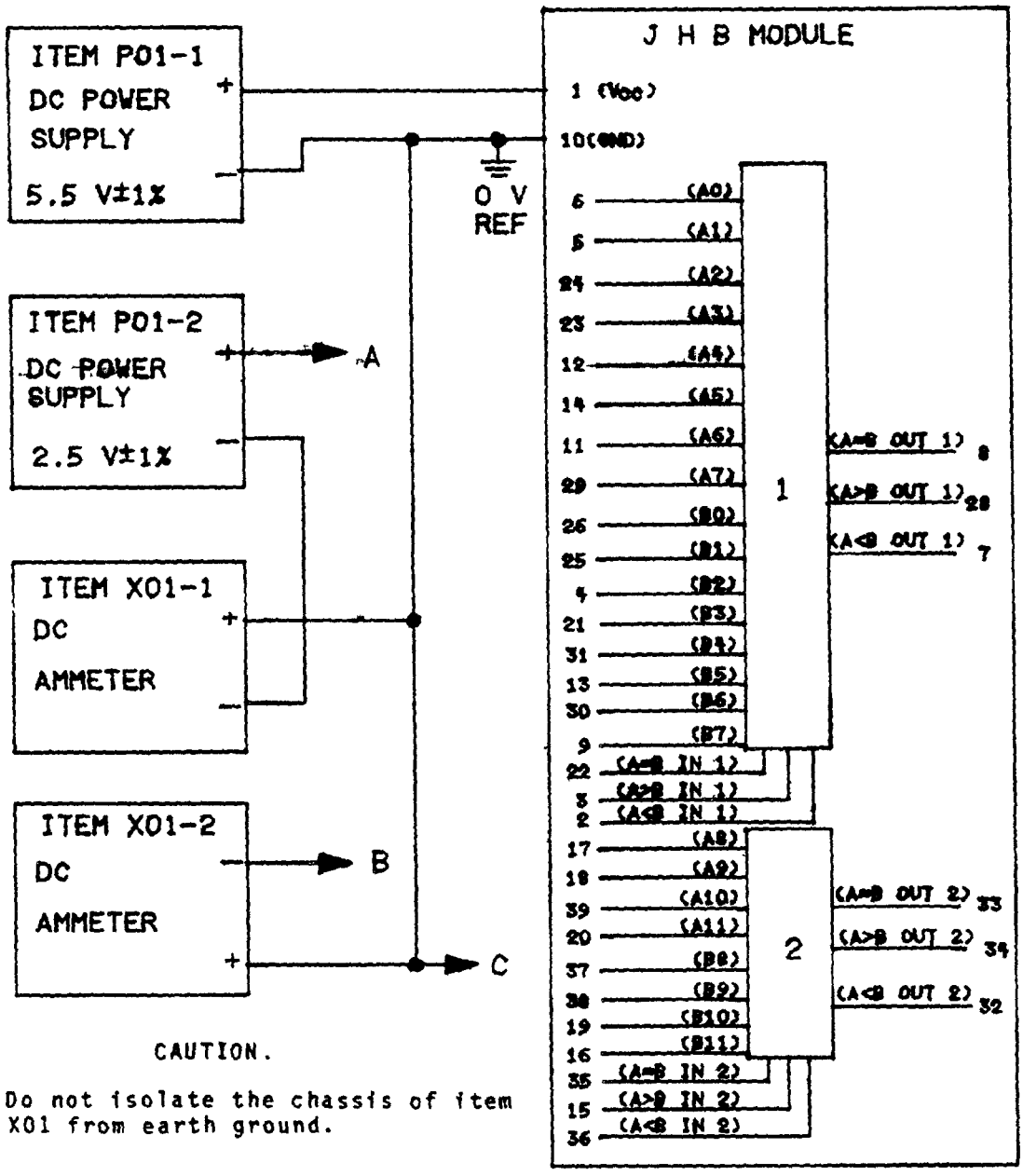
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TABLE VIII. Input current (high-level and low-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G13	B0 Input current, high level	Open	26	Open	-0.1	-6.6	mA	10
G14	B1 "	"	25	"	"	"	"	"
G15	B2 "	"	4	"	"	"	"	"
G16	B3 "	"	21	"	"	"	"	"
G17	B4 "	"	31	"	"	"	"	"
G18	B5 "	"	13	"	"	"	"	"
G19	B6 "	"	30	"	"	"	"	"
G20	B7 "	"	9	"	"	"	"	"
G21	B8 "	"	37	"	"	"	"	"
G22	B9 "	"	38	"	"	"	"	"
G23	B10 "	"	19	"	"	"	"	"
G24	B11 "	"	16	"	"	"	"	"
G25	A=B IN 1 "	"	22	"	"	"	"	"
G26	A=B IN 2 "	"	35	"	"	"	"	"
G27	A>B IN 1 "	"	3	"	"	-2.2	"	"
G28	A<B IN 1 "	"	2	"	"	"	"	"
G29	A>B IN 2 "	"	15	"	"	"	"	"
G30	A<B IN 2 "	"	36	"	"	"	"	"

1/ All input pins except the one connected to point A should be placed in the C to pin column.

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NOTE: Since items X01-1 and X01-2 are not used simultaneously, only one item X01 is required.

FIGURE 10. Input current (high-level and low-level) test circuit for table VIII.

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TABLE IX. Functional compare test.

Test condition	Parameter	Connect		Circuit 1			Circuit 2			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Pin			Pin			Min	Max		
				8	28	7	33	34	32				
H1	Functional compare	22,30,36	2-6,9,11-21,23-26,29,31,35,37-39	L	L	H	L	L	H	1/ 2/	1/ 2/	V	11
H2	"	9,15,20,22,35,36	2-6,11-14,16-19,21,23-26,29-31,37-39	L	L	H	L	H	L	"	"	"	"
H3	"	12,15,16,22,35,36	2-6,9,11,13,14,17-21,23-26,29-31,37-39	L	H	L	L	L	H	"	"	"	"
H4	"	15,22,30,35,36,39	2-6,9,11-14,16-21,23-26,29,31,37,38	L	L	H	L	H	L	"	"	"	"
H5	"	15,19,22,31,35,36	2-6,9,11-14,16-18,20,21,23-26,29,30,37-39	L	L	H	L	L	H	"	"	"	"
H6	"	13,15,18,22,35,36	2-6,9,11,12,14,16,17,19-21,23-26,29-31,37-39	L	L	H	L	H	L	"	"	"	"
H7	"	15,22,35,36,38	2-6,9,11-14,16-21,23-26,29-31,37,39	H	L	L	L	L	H	"	"	"	"
H8	"	11,15,17,22,35,36	2-6,9,12-14,16,18-21,23-26,29-31,37-39	L	H	L	L	H	L	"	"	"	"
H9	"	6,15,22,26,35,37	2-5,9,11-14,16-21,23-25,29-31,36,38,39	H	L	L	L	L	H	"	"	"	"
H10	"	9,22,35	2-6,11-21,23-26,29-31,36-39	L	L	H	H	L	L	"	"	"	"
H11	"	17,35,37	2-6,9,11-16,18-26,29-31,36,38,39	L	L	L	H	L	L	"	"	"	"
H12	"	35,38,2,3,15,18,22,23	4-6,9,11-14,16,17,19-21,24-26,29-31,36,37,38	L	H	L	H	L	L	"	"	"	"

See footnotes at end of table.

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TABLE IX. Functional compare test - Continued.

Test condition	Parameter	Connect		Circuit 1			Circuit 2			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Pin			Pin			Min	Max		
				8	28	7	33	34	32				
H13	Functional compare	35,39,2,3, 15,19,21,22	4-6,9,11-14,16-18, 20,23-26,29-31, 36-38	L	L	H	H	L	L	1/ 2/	1/ 2/	V	11
H14	"	35,2,3,15, 16,20,22,24	4-6,9,11-14,17-19, 21,23,25,26,29-31, 36-39	L	H	L	H	L	L	"	"	"	"
H15	"	2,15,22,31	3-6,9,11-14,16-21, 23-26,29,30,35-39	L	L	H	L	H	L	"	"	"	"
H16	"	4,17,24,37	2,3,5,6,9,11-16,18- 23,25,26,29-31,35, 36,38,39	L	L	L	L	H	H	"	"	"	"
H17	"	5,18,25,36, 37	26,29-31,35,38,39, 2-4,6,9,11-17,19- 24	L	L	L	L	H	L	"	"	"	"
H18	"	38,3,4,17, 18,22,36	23-26,29-31,35,37, 39,2,5,6,9,11-16, 19-21	L	L	H	L	H	L	"	"	"	"
H19	"	6,26,36-39	35,2-5,9,11-25, 29-31	L	L	L	L	H	L	"	"	"	"
H20	"	39,2,3,5, 22,37	29-31,35,36,38,4, 6,9,11-21,23-26	L	H	L	L	H	L	"	"	"	"
H21	"	25,37,39,2, 3,18,19,22	24,26,29-31,35,36, 38,4-6,9,11-17,20, 21,23	L	L	H	L	H	L	"	"	"	"
H22	"	26,37,38,2, 3,19,20,22	29-31,35,36,39,4-6, 9,11-18,21,23-25	L	L	H	L	H	L	"	"	"	"
H23	"	37,38,2,3, 6,20,22	29-31,35,36,39,4,5, 9,11-19,21,23-26	L	H	L	L	H	L	"	"	"	"
H24	"	37,2,3,20, 22,26	29-31,35,36,38,39, 4-6,9,11-19,21, 23-25	L	L	H	L	H	L	"	"	"	"

See footnotes at end of table.

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TABLE IX. Functional compare test - Continued.

Test condition	Parameter	Connect		Circuit 1			Circuit 2			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Pin			Pin			Min	Max		
				8	28	7	33	34	32				
H25	Functional compare	29,38,39,2, 3,16,20,22	23-26,30,31,35-37, 4-6,9,11-15,17-19, 21	L	H	L	L	H	L	1/ 2/	1/ 2/	V	11
H26	"	39,2,3,14, 19,22	21,23-26,29-31,35- 38,4-6,9,11-13,15- 18,20	L	H	L	L	H	H	"	"	"	"
H27	"	38,2,3,13, 18,22	23-26,29-31,35-37, 39,4-6,9,11,12,14- 17,19-21	L	L	H	L	H	H	"	"	"	"
H28	"	25,2,3,5, 16,17,22	23,24,26,29-31,35- 39,4,6,9,11-15,17- 21	H	L	L	L	L	H	"	"	"	"
H29	"	24,35,2,3, 4,17,19,22	23,25,26,29-31,36- 39,5,6,9,11-16,18, 20,21	H	L	L	L	L	H	"	"	"	"
H30	"	35,38,2,3, 17,21,22,23	25,26,29-31,36,37, 39,4-6,9,11-16,18- 20,24	H	L	L	L	L	H	"	"	"	"
H31	"	31,35,2,3, 12,16-18,22	23-26,29,30,36-39, 4-6,9,11,13-15,19- 21	H	L	L	L	L	H	"	"	"	"
H32	"	22,35,2,3, 13,14,18,19	29-31,20,21,23-26, 36-39,4,6,9,11,12 15-17	H	L	L	L	L	H	"	"	"	"
H33	"	35,39,2,3, 11,16,22,30	23-26,29,31,36-38, 4-6,9,12-15,17-21	H	L	L	L	L	H	"	"	"	"
H34	"	22,29,35,2, 3,9,16,19, 20	23-26,30,31,36-39, 4-6,11-15,17,18,21	H	L	L	L	L	H	"	"	"	"
H35	"	35,2,3,6, 16,19-22	29-31,36-39,4,5,9, 11-15,17,18,23-26	L	L	H	L	L	H	"	"	"	"

See footnotes at end of table.

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TABLE IX. Functional compare test - Continued.

Test condition	Parameter	Connect		Circuit 1			Circuit 2			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Pin			Pin			Min	Max		
				8	28	7	33	34	32				
H36	Functional compare	2-4,6,22	29-31,35-39,5,9, 11-21,23-26,	L	L	H	---	---	---	1/ 2/	1/ 2/	V	11
H37	"	2,3,6,22,25	29-31,35-39,4,5,9, 11-21,23,24,26,	L	L	H	---	---	---	"	"	"	"
H38	"	2,3,5,6,21, 22	29-31,35-39,4,9, 11-20,23-26	L	L	H	---	---	---	"	"	"	"
H39	"	2-5,22	29-31,35-39,6,9, 11-21,23-26	L	L	H	---	---	---	"	"	"	"
H40	"	2,3,21,22, 24	26,29-31,35-39,4-6, 9,11-20,23,25	L	L	H	---	---	---	"	"	"	"
H41	"	2,3,4,21, 22,23	29-31,35-39,5,6,9, 11-20,24-26	L	L	H	---	---	---	"	"	"	"
H42	"	2,3	35-39,4-6,9,11-26, 29-31	L	H	H	---	---	---	"	"	"	"
H43	"	2,12	29-31,35-39,3-6,9, 11,13-26	L	H	L	---	---	---	"	"	"	"
H44	"	2,3,12,31	30,35-39,4-6,9,11, 13-26,29	L	H	H	---	---	---	"	"	"	"
H45	"	2,14,31	29,30,35-39,3-6,9, 11-13,16-26	L	H	L	---	---	---	"	"	"	"
H46	"	2,14	29-31,35-39,3-6,9, 11-13,15-26	L	H	L	---	---	---	"	"	"	"
H47	"	2,3,13,14	29-31,35-39,4-6,9, 11,12,15-26	L	H	H	---	---	---	"	"	"	"
H48	"	2,3,11,13, 31	29,30,35-39,4-6,9, 12,14-26	L	H	L	---	---	---	"	"	"	"
H49	"	2,11,26,31	35-39,3-6,9,12-25, 29,30	L	H	L	---	---	---	"	"	"	"

See footnotes at end of table.

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TABLE IX. Functional compare test - Continued.

Test condition	Parameter	Connect		Circuit 1			Circuit 2			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Pin			Pin			Min	Max		
				8	28	7	33	34	32				
H50	Functional compare	2,11,26	35-39,3-6,9,12-25, 29-31	L	H	L	---	---	---	1/ 2/	1/ 2/	V	11
H51	"	14,2,11,26, 30,31	29,35-39,3-6,9,12, 13,15-25	L	H	L	---	---	---	"	"	"	"
H52	"	2,13,26,29-31	35-39,3-6,9,11,12, 14-25	L	H	L	---	---	---	"	"	"	"
H53	"	2,13,26,29, 31	30,35-39,3-6,9,11, 12,14-25	L	H	L	---	---	---	"	"	"	"
H54	"	2,26,29,31	35-39,3-6,9,11-25, 30	L	H	L	---	---	---	"	"	"	"
H55	"	2,26,29	35-39,3-6,9,11-25, 30,31	L	H	L	---	---	---	"	"	"	"
H56	"	29,2,5,9,11 13,26	35-39,3,4,6,12,14-25,30,31	L	H	L	---	---	---	"	"	"	"
H57	"	2,5,22,26	25,29-31,35-39,3,4, 6,9,11-21,23,24	L	H	L	---	---	---	"	"	"	"
H58	"	2,5,6,22,25	29-31,35-39,3,4,9, 11-21,23,24,26	L	H	L	---	---	---	"	"	"	"
H59	"	22,24-26	35-39,2-6,9,11-21, 23,29-31	L	H	L	---	---	---	"	"	"	"
H60	"	22,24,26	29-31,35-39,2-6,9, 11-21,23,25	L	H	L	---	---	---	"	"	"	"
H61	"	4,5,22,24, 26	29-31,35-39,2,3,6, 9,11-21,23,25	L	H	L	---	---	---	"	"	"	"
H62	"	4,22,23,25, 26	29-31,35-39,2,3,5, 6,9,11-21,24	L	H	L	---	---	---	"	"	"	"
H63	"	22,23,25,26	29-31,35-39,2-6,9, 11-21,24	L	H	L	---	---	---	"	"	"	"

See footnotes at end of table.

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TABLE IX. Functional compare test - Continued.

Test condition	Parameter	Connect		Circuit 1			Circuit 2			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Pin			Pin			Min	Max		
				8	28	7	33	34	32				
H64	Functional compare	22,23,26	25,29-31,35-39,2-6, 9,11-21,24	L	H	L	---	---	---	1/ 2/	1/ 2/	V	11
H65	"	21,22,23, 24,25	29-31,35-39,2-6,9, 11-20,26	L	H	L	---	---	---	"	"	"	"
H66	"	11,21,22,30	29,31,35-39,2-6,9, 12-20,23-26	L	L	H	---	---	---	"	"	"	"
H67	"	13,14,21,22	23-26,29-31,35-39, 2-6,9,11,12,15-20	L	L	H	---	---	---	"	"	"	"
H68	"	12,21,22,31	29,30,35-39,2-6,9, 11,13-20,23-26	L	L	H	---	---	---	"	"	"	"
H69	"	9	35-39,2-6,11-26, 29-31	L	L	H	---	---	---	"	"	"	"
H70	"	30	9,35-39,2-6,11-26, 29,31	L	L	H	---	---	---	"	"	"	"
H71	"	13	29-31,35-39,2-6,9, 11,12,14-26	L	L	H	---	---	---	"	"	"	"
H72	"	31	35-39,2-6,9,11-26, 29,30	L	L	H	---	---	---	"	"	"	"
H73	"	9,12	35-39,2-6,11,13-26, 29-31	L	L	H	---	---	---	"	"	"	"
H74	"	12,21,22,30	29,31,35-39,2-6,9, 11,13-20,23-26	L	L	H	---	---	---	"	"	"	"
H75	"	12,13,21,22	23-26,29-31,35-39, 2-6,9,11,14-20	L	L	H	---	---	---	"	"	"	"
H76	"	9,12,14,21, 22	23-26,29-31,35-39, 2-6,11,13,15-20	L	L	H	---	---	---	"	"	"	"

See footnotes at end of table.

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TABLE IX. Functional compare test - Continued.

Test condition	Parameter	Connect		Circuit 1			Circuit 2			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Pin			Pin			Min	Max		
				8	28	7	33	34	32				
H77	Functional compare	14,21,22,30	23-26,29,31,35-39, 2-6,9,11-13,15-20	L	L	H	---	---	---	1/ 2/	1/ 2/	V	11
H78	"	6,9,21,22	29-31,35-39,2-5, 11-20,23-26	L	L	H	---	---	---	"	"	"	"
H79	"	6,21,22,29	30,31,35-39,2-5, 9,11-20,23-26	L	L	H	---	---	---	"	"	"	"

1/ Output logic states:
L = low level ≤ 500 mV.
H = high level ≥ 2.5 V.

2/ For test condition H, verification of the output states specified in footnote 1 is obtained by connecting item M02 (point C) to each output pin after input conditions are accomplished.

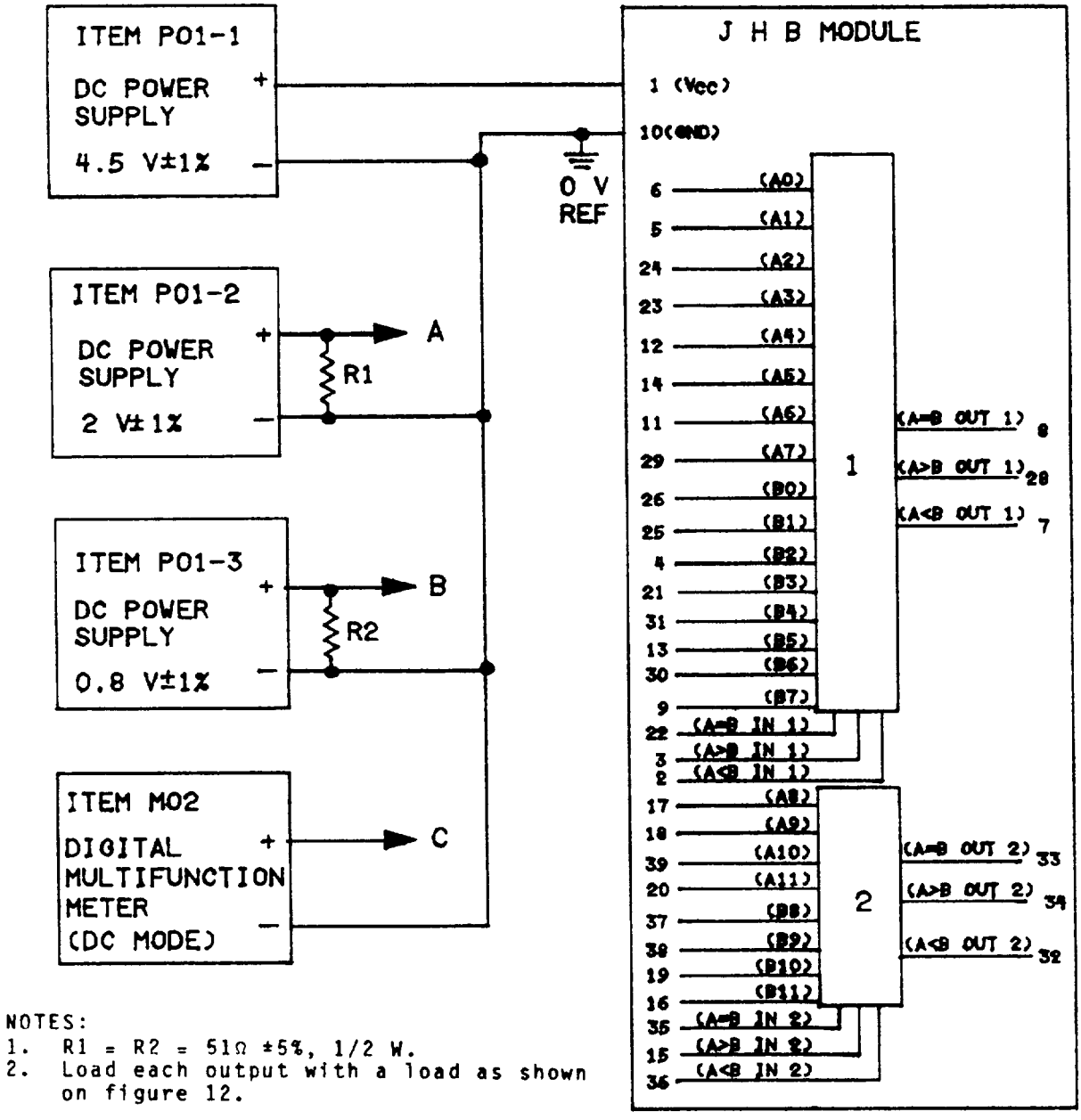
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TABLE X Output voltage (high-level and low-level) test.

Test condition	Parameter		Connect			Outputs 1/			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	A=B	A>B	A<B	Min	Max		
J1	A=B OUT 1	Output voltage, high level	22	2,3,4-6,9,11-14,21, 23-26,29-31	8	H	L	L	2.5	4.5	V	11
J2	A>B OUT 1	"	3	2,22,4-6,9,11-14,21, 23-26,29-31	28	L	H	L	"	"	"	"
J3	A<B OUT 1	"	2	3,22,4-6,9,11-14,21, 23-26,29-31	7	L	L	H	"	"	"	"
J4	A=B OUT 2	"	35	15,36,16-20,37,38,39	33	H	L	L	"	"	"	"
J5	A>B OUT 2	"	15	35,36,16-20,37,38,39	34	L	H	L	"	"	"	"
J6	A<B OUT 2	"	36	15,35,16-20,37,38,39	32	L	L	H	"	"	"	"
K1	A=B OUT 1	Output voltage, low level	2	22,3,4-6,9,11-14,21, 23-26,29-31	8	L	L	H	0	500	mV	"
K2	A>B OUT 1	"	22	3,2,4-6,9,11-14,21, 23-26,29-31	28	H	L	L	"	"	"	"
K3	A<B OUT 1	"	3	2,22,4-6,9,11-14,21, 23-26,29-31	7	L	H	L	"	"	"	"
K4	A=B OUT 2	"	36	15,35,16-20,37,38,39	33	L	L	H	"	"	"	"
K5	A>B OUT 2	"	35	15,36,16-20,37,38,39	34	H	L	L	"	"	"	"
K6	A<B OUT 2	"	15	35,36,16-20,37,38,39	32	L	H	L	"	"	"	"

1/ Output logic states.
L = low level ≤ 500 mV.
H = high level ≥ 2.5 V.

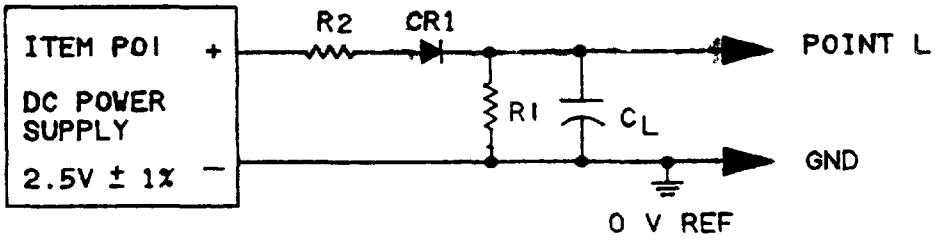
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. Load each output with a load as shown on figure 12.

FIGURE 11. Functional compare and output voltage (high-level and low-level) test circuit for tables IX and X.

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NOTES:
1. CR1 = 1N4150, 1N3600 or equivalent.
2. R1 = 2.49 kΩ ±1%, 1/8 W.
3. R2 = Select value for a current flow of 20 mA ±1% out of point L with point L held at 0.5 V.
4. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions M1-M106, N1-N106, Q1-Q6, and R1-R6.
5. C_L is not required for test conditions H1-H79, J1-J6, and K1-K6.
6. Lead length from load circuit to module output pin under test shall be less than 2 inches (50.8 mm).

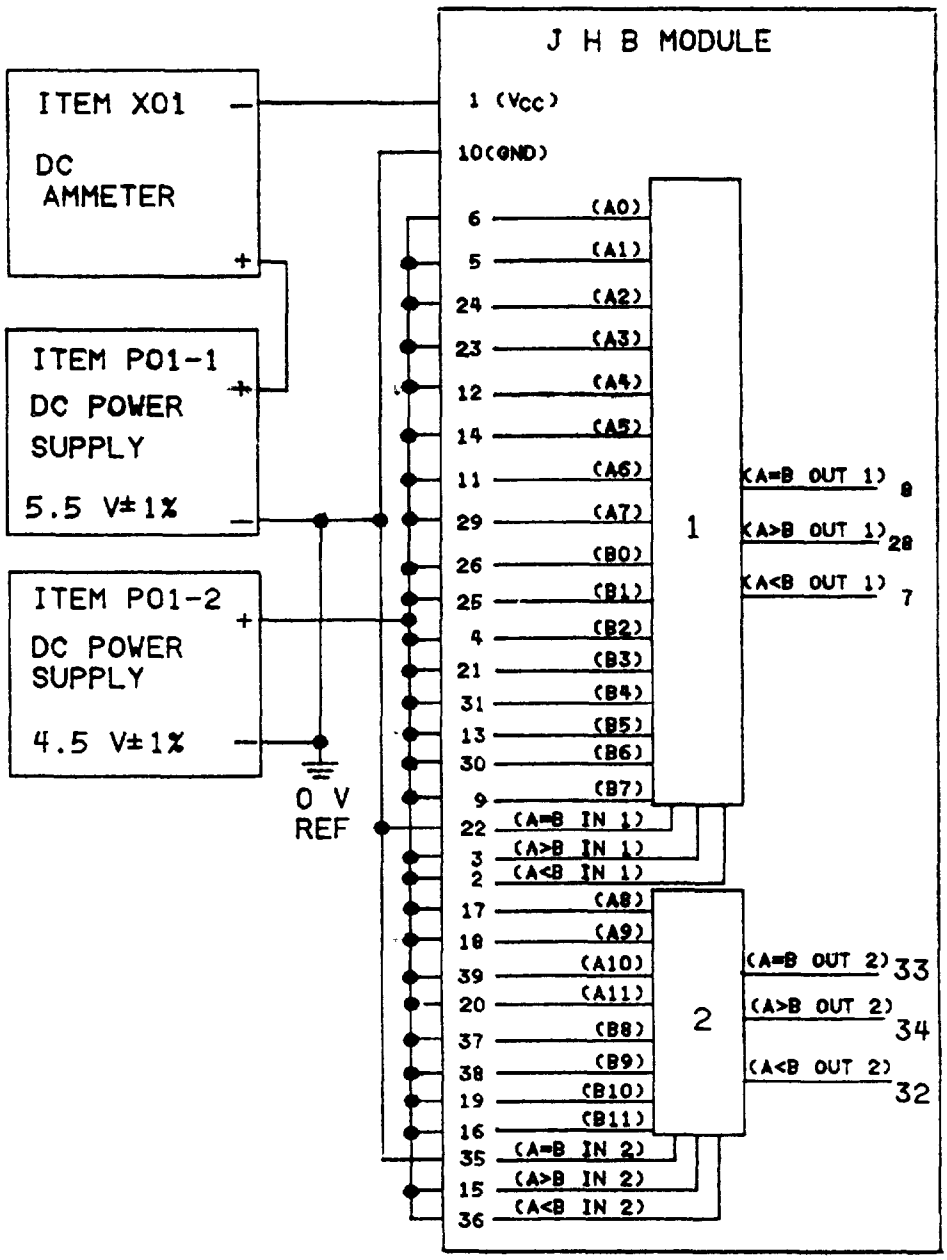
FIGURE 12. JHB logic load circuit.

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TABLE XI. Power supply current test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
L	Power supply current	30	345	mA	13

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CAUTION:

Do not isolate the chassis of item XO1 from earth ground.

FIGURE 13. Power supply current test circuit for table XI.

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TABLE XII Propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
M1	A<B to A>B OUT 1 Propagation delay time (t_{PLH}), C_L = 50 pF	Open	2	3	28	4,5,6,9,11,12,13,14,21, 22,23,24,25,26,29,30,31	2	19	ns	14,15
M2	A=B to A>B OUT 1 "	"	22	3	28	2	"	"	"	"
M3	A>B to A<B OUT 1 "	"	3	2	7	22	"	"	"	"
M4	A=B to A<B OUT 1 "	"	22	"	7	3	"	"	"	"
M5	A=B to A=B OUT 1 "	"	22	"	8	3,4,5,6,9,11,12,13,14, 21,23,24,25,26,29,30,31	"	26	"	"
M6	A<B to A>B OUT 2 "	"	36	15	34	16,17,18,19,20,35,37, 38,39	2.0	9.5	"	"
M7	A=B to A>B OUT 2 "	"	35	15	34	36	"	"	"	"
M8	A>B to A<B OUT 2 "	"	15	36	32	35	"	"	"	"
M9	A=B to A<B OUT 2 "	"	35	"	32	15	"	"	"	"
M10	A=B to A=B OUT 2 "	"	35	"	33	15,16,17,18,19,20,37, 38,39	2	13	"	"
M11	A11>B11, pulse on A, 0 V on B "	"	20	Open	34	16	"	19	"	"
M12	A10>B10, pulse on A, 0 V on B "	"	39	"	"	16,19,20	"	"	"	"
M13	A9>B9, pulse on A, 0 V on B "	"	18	"	"	16,19,20,38,39	"	"	"	"
M14	A8>B8, pulse on A, 0 V on B "	"	17	"	"	16,18,19,20,37,38,39	"	"	"	"
M15	A11=B11, pulse on A, 0 V on B "	"	20	"	33	16	"	21	"	"
M16	A10=B10, pulse on A, 0 V on B "	"	39	"	"	16,19,20	"	"	"	"
M17	A9=B9, pulse on A, 0 V on B "	"	18	"	"	16,19,20,38,39	"	"	"	"
M18	A8=B8, pulse on A, 0 V on B "	"	17	"	"	16,18,19,20,37,38,39	"	"	"	"

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
M19	A11<B11 Propagation delay time (t_{PLH}), C_L = 50 pF, pulse on A, 4.5 V on B	16	20	Open	32	Open	2	19	ns	14,15
M20	A10<B10 "	16,19,20	39	"	"	"	"	"	"	"
M21	A9<B9 "	16,19,20, 38,39	18	"	"	"	"	"	"	"
M22	A8<B8 "	16,18,19,20, 37,38,39	17	"	"	"	"	"	"	"
M23	A11=B11 "	16	20	"	33	"	"	21	"	"
M24	A10=B10 "	16,19,20	39	"	"	"	"	"	"	"
M25	A9=B9 "	16,19,20, 38,39	18	"	"	"	"	"	"	"
M26	A8=B8 "	16,18,19,20, 37,38,39	17	"	"	"	"	"	"	"
M27	A11>B11 Propagation delay time (t_{PLH}), C_L = 50 pF, pulse on B, 4.5 V on A	20	16	"	34	"	"	19	"	"
M28	A10>B10 "	16,20,39	19	"	"	"	"	"	"	"
M29	A9>B9 "	16,18,19, 20,39	38	"	"	"	"	"	"	"
M30	A8>B8 "	16,17,18,19, 20,38,39	37	"	"	"	"	"	"	"
M31	A11=B11 "	20	16	"	33	"	"	21	"	"
M32	A10=B10 "	16,20,39	19	"	"	"	"	"	"	"
M33	A9=B9 "	16,18,19, 20,39	38	"	"	"	"	"	"	"
M34	A8=B8 "	16,17,18,19, 20,38,39	37	"	"	"	"	"	"	"

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
M35	A11<B11 Propagation delay time (t_{PLH}), C_L = 50 pF, pulse on B, 0 V on A	Open	16	Open	32	20	2	19	ns	14,15
M36	A10<B10 "	"	19	"	"	16,20,39	"	"	"	"
M37	A9<B9 "	"	38	"	"	16,18,19,20,39	"	"	"	"
M38	A8<B8 "	"	37	"	"	16,17,18,19,20,38,39	"	"	"	"
M39	A11=B11 "	"	16	"	33	20	"	21	"	"
M40	A10=B10 "	"	19	"	"	16,20,39	"	"	"	"
M41	A9=B9 "	"	38	"	"	16,18,19,20,39	"	"	"	"
M42	A8=B8 "	"	37	"	"	16,17,18,19,20,38,39	"	"	"	"
M43	A7>B7 Propagation delay time (t_{PLH}), C_L = 50 pF, pulse on A, 0 V on B	"	29	"	28	9	"	19	"	"
M44	A6>B6 "	"	11	"	"	9,29,30	"	"	"	"
M45	A5>B5 "	"	14	"	"	9,11,13,29,30	"	"	"	"
M46	A4>B4 "	"	12	"	"	9,11,13,14,29,30,31	"	"	"	"
M47	A3>B3 "	"	23	"	"	9,11,12,13,14,21,29,30,31	2.0	28.5	"	"
M48	A2>B2 "	"	24	"	"	4,9,11,12,13,14,21,23, 29,30,31	"	"	"	"
M49	A1>B1 "	"	5	"	"	4,9,11,12,13,14,21,23, 24,25,29,30,31	"	"	"	"
M50	A0>B0 "	"	6	"	"	4,5,9,11,12,13,14,21,23, 24,25,26,29,30,31	"	"	"	"
M51	A7=B7 "	"	29	"	8	9	2	21	"	"
M52	A6=B6 "	"	11	"	8	9,29,30	2	21	"	"

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
M53	A5=B5	Propagation delay time (t_{PLH}), C_L = 50 pF, pulse on A, 0 V on B	Open	14	Open	8	9,11,13,29,30	2	21	ns	14,15
M54	A4=B4	"	"	12	"	"	9,11,13,14,29,30,31	"	21	"	"
M55	A3=B3	"	"	23	"	"	9,11,12,13,14,21,29, 30,31	"	34	"	"
M56	A2=B2	"	"	24	"	"	4,9,11,12,13,14,21,23, 29,30,31	"	"	"	"
M57	A1=B1	"	"	5	"	"	4,9,11,12,13,14,21,23, 24,25,29,30,31	"	"	"	"
M58	A0=B0	"	"	6	"	"	4,5,9,11,12,13,14,21,23, 24,25,26,29,30,31	"	"	"	"
M59	A7<B7	Propagation delay time (t_{PLH}), C_L = 50 pF, pulse on A, 4.5 V on B	9	29	"	7	Open	"	19	"	"
M60	A6<B6	"	9,29,30	11	"	"	"	"	"	"	"
M61	A5<B5	"	9,11,13, 29,30	14	"	"	"	"	"	"	"
M62	A4<B4	"	9,11,13,14, 29,30,31	12	"	"	"	"	"	"	"
M63	A3<B3	"	9,11,12,13, 14,21,29,30, 31	23	"	"	"	2.0	28.5	"	"
M64	A2<B2	"	4,9,11,12, 13,14,21,23, 29,30,31	24	"	"	"	"	"	"	"
M65	A1<B1	"	4,9,11,12, 13,14,21,23, 24,25,29,30, 31	5	"	"	"	"	"	"	"

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
M66	A0<B0	Propagation delay time (t_{PLH}), C_L = 50 pF, pulse on A, 4.5 V on B	4,5,9,11,12, 13,14,21,23, 24,25,26,29, 30,31	6	Open	7	Open	2.0	28.5	ns	14,15
M67	A7=B7	"	9	29	"	8	"	2	21	"	"
M68	A6=B6	"	9,29,30	11	"	"	"	"	"	"	"
M69	A5=B5	"	9,11,13, 29,30	14	"	"	"	"	"	"	"
M70	A4=B4	"	9,11,13,14, 29,30,31	12	"	"	"	"	"	"	"
M71	A3=B3	"	9,11,12,13, 14,21,29,30, 31	23	"	"	"	"	34	"	"
M72	A2=B2	"	4,9,11,12, 13,14,21,23, 29,30,31	24	"	"	"	"	"	"	"
M73	A1=B1	"	4,9,11,12, 13,14,21,23, 24,25,29,30, 31	5	"	"	"	"	"	"	"
M74	A0=B0	"	4,5,9,11,12, 13,14,21,23, 24,25,26,29, 30,31	6	"	"	"	"	"	"	"
M75	A7>B7	Propagation delay time (t_{PLH}), C_L = 50 pF, pulse on B, 4.5 V on A	29	9	"	28	"	"	19	"	"
M76	A6>B6	"	9,11,29	30	"	"	"	"	"	"	"
M77	A5>B5	"	9,11,14, 29,30	13	"	"	"	"	"	"	"

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition		Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
M78	A4>B4	Propagation delay time (t_{PLH}), C_L = 50 pF, pulse on B, 4.5 V on A	9,11,12,13, 14,29,30	31	Open	28	Open	2	19	ns	14,15
M79	A3>B3	"	9,11,12,13, 14,23,29,30, 31	21	"	"	"	2.0	28.5	"	"
M80	A2>B2	"	9,11,12,13, 14,21,23,24, 29,30,31	4	"	"	"	"	"	"	"
M81	A1>B1	"	4,5,9,11,12, 13,14,21,23, 24,29,30,31	25	"	"	"	"	"	"	"
M82	A0>B0	"	4,5,6,9,11, 12,13,14,21, 23,24,25,29, 30,31	26	"	"	"	"	"	"	"
M83	A7=B7	"	29	9	"	8	"	2	21	"	"
M84	A6=B6	"	9,11,29	30	"	"	"	"	"	"	"
M85	A5=B5	"	9,11,14, 29,30	13	"	"	"	"	"	"	"
M86	A4=B4	"	9,11,12,13, 14,29,30	31	"	"	"	"	"	"	"
M87	A3=B3	"	9,11,12,13, 14,23,29,30, 31	21	"	"	"	"	34	"	"
M88	A2=B2	"	9,11,12,13, 14,23,24,29, 30,31	4	"	"	"	"	"	"	"
M89	A1=B1	"	4,5,9,11,12, 13,14,21,23, 24,29,30,31	25	"	"	"	"	"	"	"

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
M90	A0>B0	Propagation delay time (t_{PLH}), C_L = 50 pF, pulse on B, 4.5 V on A	4,5,6,9,11, 12,13,14,21, 23,24,25,29, 30,31	26	Open	8	Open	2	34	ns	14,15
M91	A7<B7	Propagation delay time (t_{PLH}), C_L = 50 pF, pulse on B, 0 V on A	Open	9	"	7	29	"	19	"	"
M92	A6<B6	"	"	30	"	"	9,11,29	"	"	"	"
M93	A5<B5	"	"	13	"	"	9,11,14,29,30	"	"	"	"
M94	A4<B4	"	"	31	"	"	9,11,12,13,14,29,30	"	"	"	"
M95	A3<B3	"	"	21	"	"	9,11,12,13,14,23,29, 30,31	2.0	28.5	"	"
M96	A2<B2	"	"	4	"	"	9,11,12,13,14,21,23,24, 29,30,31	"	"	"	"
M97	A1<B1	"	"	25	"	"	4,5,9,11,12,13,14,21,23, 24,29,30,31	"	"	"	"
M98	A0<B0	"	"	26	"	"	4,5,6,9,11,12,13,14,21, 23,24,25,29,30,31	"	"	"	"
M99	A7=B7	"	"	9	"	8	29	2	21	"	"
M100	A6=B6	"	"	30	"	"	9,11,29	"	"	"	"
M101	A5=B5	"	"	13	"	"	9,11,14,29,30	"	"	"	"
M102	A4=B4	"	"	31	"	"	9,11,12,13,14,29,30	"	"	"	"
M103	A3=B3	"	"	21	"	"	9,11,12,13,14,23,29, 30,31	"	34	"	"
M104	A2=B2	"	"	4	"	"	9,11,12,13,14,21,23,24, 29,30,31	"	"	"	"
M105	A1=B1	"	"	25	"	"	4,5,9,11,12,13,14,21,23, 24,29,30,31	"	"	"	"

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
M106	AO=BO Propagation delay time (t_{PLH}), C_L = 50 pF, pulse on B, 0 V on A	Open	26	Open	7	4,5,6,9,11,12,13,14,21, 23,24,25,29,30,31	2	34	ns	14,15
N1	A<B to A>B OUT 1 Propagation delay time (t_{PHL}), C_L = 50 pF	"	2	3	28	4,5,6,9,11,12,13,14,21, 22,23,24,25,26,29,30,31	"	21	"	"
N2	A=B to A>B OUT 1 "	"	22	3	28	2	"	"	"	"
N3	A>B to A<B OUT 1 "	"	3	2	7	22	"	"	"	"
N4	A=B to A<B OUT 1 "	"	22	"	7	3	"	"	"	"
N5	A=B to A=B OUT 1 "	"	22	"	8	3,4,5,6,9,11,12,13,14, 21,23,24,25,26,29,30,31	"	19	"	"
N6	A<B to A>B OUT 2 "	"	36	15	34	16,17,18,19,20,35,37, 38,39	2.0	10.5	"	"
N7	A=B to A>B OUT 2 "	"	35	15	34	36	"	"	"	"
N8	A>B to A<B OUT 2 "	"	15	36	32	35	"	"	"	"
N9	A=B to A<B OUT 2 "	"	35	"	32	15	"	"	"	"
N10	A=B to A=B OUT 2 "	"	35	"	33	15,16,17,18,19,20,37, 38,39	"	9.5	"	"
N11	A11>B11 Propagation delay time (t_{PHL}), C_L = 50 pF, pulse on A, 0 V on B	"	20	Open	34	16	"	19.5	"	"
N12	A10>B10 "	"	39	"	"	16,19,20	"	"	"	"
N13	A9>B9 "	"	18	"	"	16,19,20,38,39	"	"	"	"
N14	A8>B8 "	"	17	"	"	16,18,19,20,37,38,39	"	"	"	"

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
N15	A11=B11 Propagation delay time (t_{PHL}), C_L = 50 pF, pulse on A, 0 V on B	Open	20	Open	33	16	2 0	19.5	ns	14,15
N16	A10=B10 "	"	39	"	"	16,19,20	"	"	"	"
N17	A9=B9 "	"	18	"	"	16,19,20,38,39	"	"	"	"
N18	A8=B8 "	"	17	"	"	16,18,19,20,37,38,39	"	"	"	"
N19	A11<B11 Propagation delay time (t_{PHL}), C_L = 50 pF, pulse on A, 4.5 V on B	16	20	"	32	Open	"	"	"	"
N20	A10<B10 "	16,19,20	39	"	"	"	"	"	"	"
N21	A9<B9 "	16,19,20 38,39	18	"	"	"	"	"	"	"
N22	A8<B8 "	16,18,19,20 37,38,39	17	"	"	"	"	"	"	"
N23	A11=B11 "	16	20	"	33	"	"	"	"	"
N24	A10=B10 "	16,19,20	39	"	"	"	"	"	"	"
N25	A9=B9 "	16,19,20 38,39	18	"	"	"	"	"	"	"
N26	A8=B8 "	16,18,19,20, 37,38,39	17	"	"	"	"	"	"	"
N27	A11>B11 Propagation delay time (t_{PHL}), C_L = 50 pF, pulse on B, 4.5 V on A	20	16	"	34	"	"	"	"	"
N28	A10>B10 "	16,20,39	19	"	"	"	"	"	"	"
N29	A9>B9 "	16,18,19, 20,39	38	"	"	"	"	"	"	"
N30	A8=B8 "	16,17,18,19, 20,38,39	37	"	"	"	"	"	"	"

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
N31	A11=B11	Propagation delay time (t_{PHL}), C_L = 50 pF, pulse on B, 4.5 V on A	20	16	Open	33	Open	2.0	19.5	ns	14,15
N32	A10=B10	"	16,20,39	19	"	"	"	"	"	"	"
N33	A9=B9	"	16,18,19, 20,39	38	"	"	"	"	"	"	"
N34	A8=B8	"	16,17,18,19, 20,38,39	37	"	"	"	"	"	"	"
N35	A11<B11	Propagation delay time (t_{PHL}), C_L = 50 pF, pulse on B, 0 V on A	Open	16	"	32	20	"	"	"	"
N36	A10<B10	"	"	19	"	"	16,20,39	"	"	"	"
N37	A9<B9	"	"	38	"	"	16,18,19,20,39	"	"	"	"
N38	A8<B8	"	"	37	"	"	16,17,18,19,20,38,39	"	"	"	"
N39	A11=B11	"	"	16	"	33	20	"	"	"	"
N40	A10=B10	"	"	19	"	"	16,20,39	"	"	"	"
N41	A9=B9	"	"	38	"	"	16,18,19,20,39	"	"	"	"
N42	A8=B8	"	"	37	"	"	16,17,18,19,20,38,39	"	"	"	"
N43	A7>B7	Propagation delay time (t_{PHL}), C_L = 50 pF, pulse on A, 0 V on B	"	29	"	28	9	"	"	"	"
N44	A6>B6	"	"	11	"	"	9,29,30	"	"	"	"
N45	A5>B5	"	"	14	"	"	9,11,13,29,30	"	"	"	"
N46	A4>B4	"	"	12	"	"	9,11,13,14,29,30,31	"	"	"	"
N47	A3>B3	"	"	23	"	"	9,11,12,13,14,21,29, 30,31	2	30	"	"

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
N48	A2>B2	Propagation delay time (t_{PHL}), C_L = 50 pF, pulse on A, 0 V on B	Open	24	Open	28	4,9,11,12,13,14,21,23, 29,30,31	2	30	ns	14,15
N49	A1>B1	"	"	5	"	"	4,9,11,12,13,14,21,23, 24,25,29,30,31	"	"	"	"
N50	A0>B0	"	"	6	"	"	4,5,9,11,12,13,14,21,23, 24,25,26,29,30,31	"	"	"	"
N51	A7=B7	"	"	29	"	8	9	2.0	19.5	"	"
N52	A6=B6	"	"	11	"	"	9,29,30	"	"	"	"
N53	A5=B5	"	"	14	"	"	9,11,13,29,30	"	"	"	"
N54	A4=B4	"	"	12	"	"	9,11,13,14,29,30,31	"	"	"	"
N55	A3=B3	"	"	23	"	"	9,11,12,13,14,21,29, 30,31	2	29	"	"
N56	A2=B2	"	"	24	"	"	4,9,11,12,13,14,21,23, 29,30,31	"	"	"	"
N57	A1=B1	"	"	5	"	"	4,9,11,12,13,14,21,23, 24,25,29,30,31	"	"	"	"
N58	A0=B0	"	"	6	"	"	4,5,9,11,12,13,14,21,23, 24,25,26,29,30,31	"	"	"	"
N59	A7<B7	Propagation delay time (t_{PHL}), C_L = 50 pF, pulse on A, 4.5 V on B	9	29	"	7	Open	2.0	19.5	"	"
N60	A6<B6	"	9,29,30	11	"	"	"	"	"	"	"
N61	A5<B5	"	9,11,13, 29,30	14	"	"	"	"	"	"	"
N62	A4<B4	"	9,11,13,14, 29,30,31	12	"	"	"	"	"	"	"

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TABLE XII Propagation delay time (t_{PLH} and t_{PHL}) test - Continued

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
N63	A3<B3	Propagation delay time (t_{PHL}), C_L = 50 pF, pulse on A, 4.5 V on B	9,11,12,13, 14,21,29,30, 31	23	Open	7	Open	2	30	ns	14,15
N64	A2<B2	"	4,9,11,12, 13,14,21,23, 29,30,31	24	"	"	"	"	"	"	"
N65	A1<B1	"	4,9,11,12, 13,14,21,23, 24,25,29,30, 31	5	"	"	"	"	"	"	"
N66	A0<B0	"	4,5,9,11,12, 13,14,21,23, 24,25,26,29, 30,31	6	"	"	"	"	"	"	"
N67	A7=B7	"	9	29	"	8	"	2.0	19.5	"	"
N68	A6=B6	"	9,29,30	11	"	"	"	"	"	"	"
N69	A5=B5	"	9,11,13, 29,30	14	"	"	"	"	"	"	"
N70	A4=B4	"	9,11,13,14, 29,30,31	12	"	"	"	"	"	"	"
N71	A3=B3	"	9,11,12,13, 14,21,29,30, 31	23	"	"	"	2	29	"	"
N72	A2=B2	"	4,9,11,12, 13,14,21,23, 29,30,31	24	"	"	"	"	"	"	"
N73	A1=B1	"	4,9,11,12, 13,14,21,23, 24,25,29,30, 31	5	"	"	"	"	"	"	"
N74	A0=B0	"	4,5,9,11,12, 13,14,21,23, 24,25,26,29, 30,31	6	"	"	"	"	"	"	"

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
N75	A7>B7	Propagation delay time (t_{PHL}), C_L = 50 pF, pulse on B, 4.5 V on A	29	9	Open	28	Open	2.0	19.5	ns	14,15
N76	A6>B6	"	9,11,29	30	"	"	"	"	"	"	"
N77	A5>B5	"	9,11,14, 29,30	13	"	"	"	"	"	"	"
N78	A4>B4	"	9,11,12,13, 14,29,30	31	"	"	"	"	"	"	"
N79	A3>B3	"	9,11,12,13, 14,23,29,30, 31	21	"	"	"	2	30	"	"
N80	A2>B2	"	9,11,12,13, 14,21,23,24, 29,30,31	4	"	"	"	"	"	"	"
N81	A1>B1	"	4,5,9,11,12, 13,14,21,23, 24,29,30,31	25	"	"	"	"	"	"	"
N82	A0>B0	"	4,5,6,9,11, 12,13,14,21, 23,24,25,29, 30,31	26	"	"	"	"	"	"	"
N83	A7=B7	"	29	9	"	8	"	2.0	19.5	"	"
N84	A6=B6	"	9,11,29	30	"	"	"	"	"	"	"
N85	A5=B5	"	9,11,14, 29,30	13	"	"	"	"	"	"	"
N86	A4=B4	"	9,11,12,13, 14,29,30	31	"	"	"	"	"	"	"
N87	A3=B3	"	9,11,12,13, 14,23,29,30, 31	21	"	"	"	2	29	"	"

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TABLE XII Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

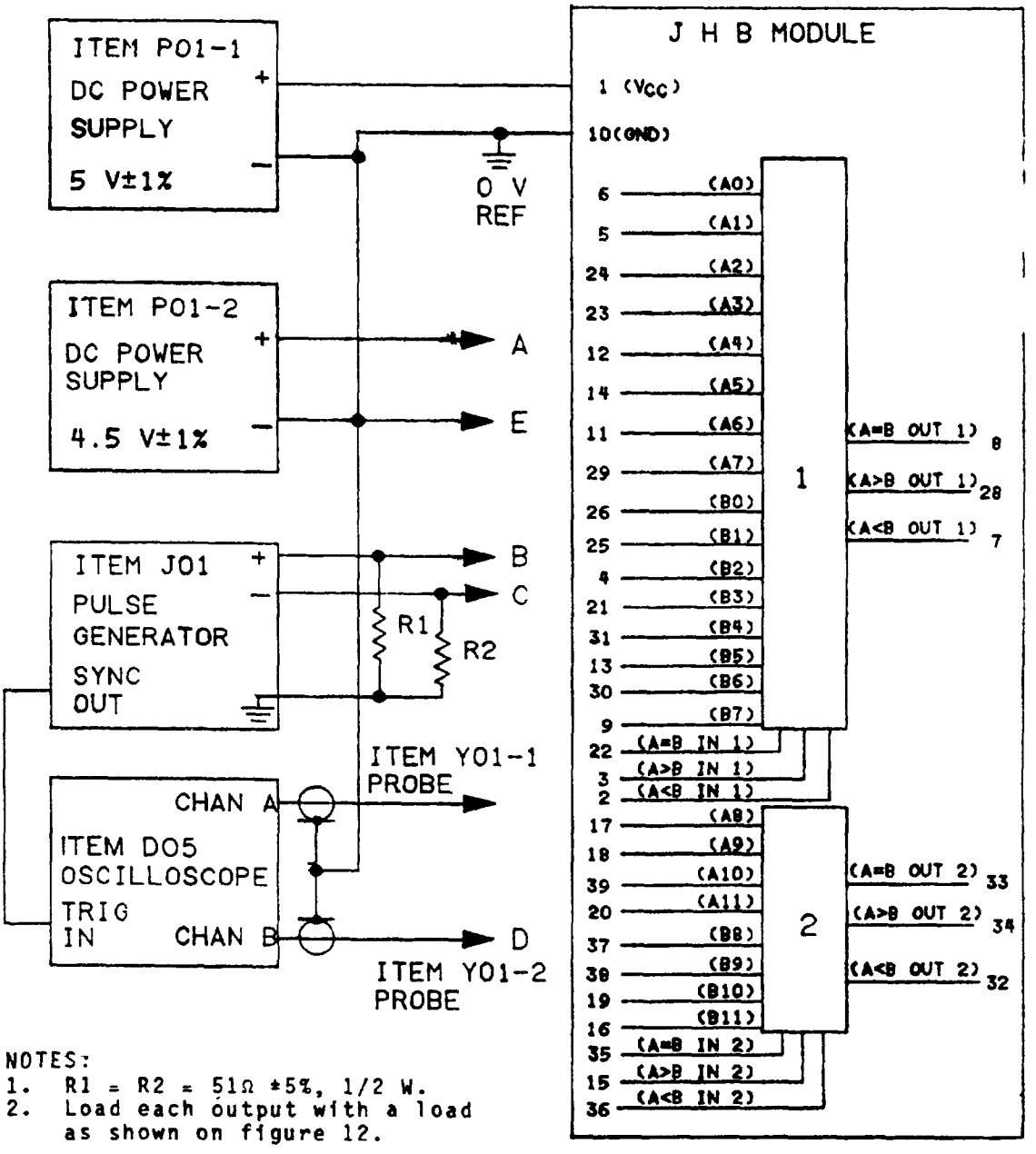
Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
N88	A2=B2	Propagation delay time (t_{PHL}), C_L = 50 pF, pulse on B, 4.5 V on A	9,11,12,13, 14,21,23,24, 29,30,31	4	Open	8	Open	2	29	ns	14,15
N89	A1=B1	"	4,5,9,11,12, 13,14,21,23, 24,29,30,31	25	"	"	"	"	"	"	"
N90	A0=B0	"	4,5,6,9,11, 12,13,14,21, 23,24,25,29, 30,31	26	"	"	"	"	"	"	"
N91	A7<B7	Propagation delay time (t_{PHL}), C_L = 50 pF, pulse on B, 0 V on A	Open	9	"	7	29	2.0	19.5	"	"
N92	A6<B6	"	"	30	"	"	9,11,29	"	"	"	"
N93	A5<B5	"	"	13	"	"	9,11,14,29,30	"	"	"	"
N94	A4<B4	"	"	31	"	"	9,11,12,13,14,29,30	"	"	"	"
N95	A3<B3	"	"	21	"	"	9,11,12,13,14,23,29, 30,31	2	30	"	"
N96	A2<B2	"	"	4	"	"	9,11,12,13,14,21,23,24, 29,30,31	"	"	"	"
N97	A1<B1	"	"	25	"	"	4,5,9,11,12,13,14,21,23, 24,29,30,31	"	"	"	"
N98	A0<B0	"	"	26	"	"	4,5,6,11,12,13,14,21,23, 24,25,29,30,31,9	"	"	"	"
N99	A7=B7	"	"	9	"	8	29	2.0	19.5	"	"
N100	A6=B6	"	"	30	"	"	9,11,29	"	"	"	"
N101	A5=B5	"	"	13	"	"	9,11,14,29,30	"	"	"	"
N102	A4=B4	"	"	31	"	"	9,11,12,13,14,29,30	"	"	"	"

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TABLE XII Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
N103	A3=B3 Propagation delay time (t_{PHL}), C_L = 50 pF, pulse on B, 0 V on A	Open	21	Open	8	9,11,12,13,14,23,29, 30,31	2	29	ns	14,15
N104	A2=B2 "	"	4	"	"	9,11,12,13,14,21,23,24, 29,30,31	"	"	"	"
N105	A1=B1 "	"	25	"	"	4,5,9,11,12,13,14,21,23, 24,29,30,31	"	"	"	"
N106	A0=B0 "	"	26	"	"	4,5,6,9,11,12,13,14,21, 23,24,25,29,30,31	"	"	"	"

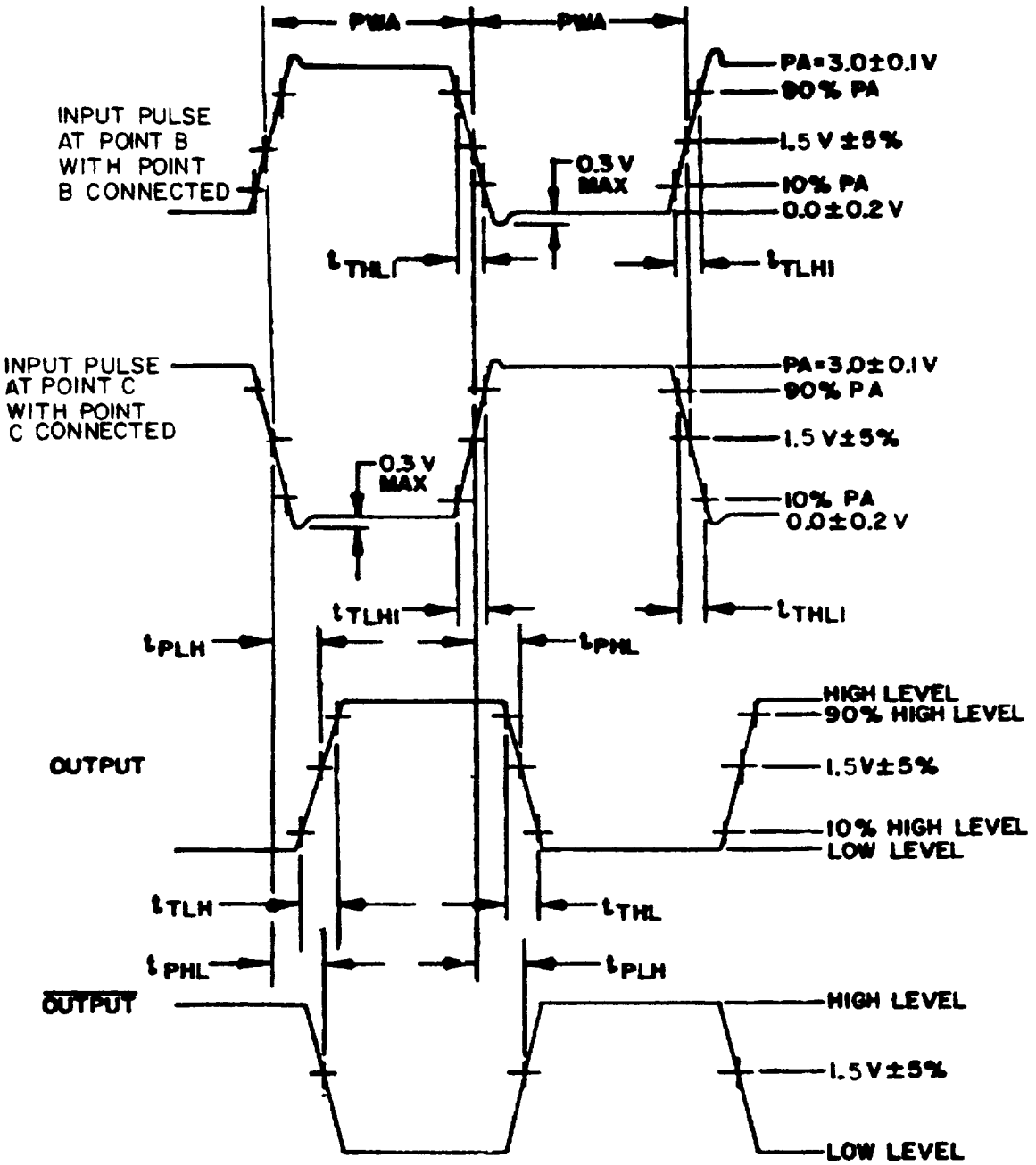
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. Load each output with a load as shown on figure 12.

FIGURE 14. Propagation delay time (t_{PLH} and t_{PHL}) test circuit for table XII.

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NOTES:
1. Input pulse parameters:
   PWA = 500 ±5 ns.
   $t_{TLH1} = t_{THL1}$ = 6 ±2 ns.

2. Before making measurements, the output levels shall be verified to be $\leq$0.5 V for the low level and $\geq$2.5 V for the high level.

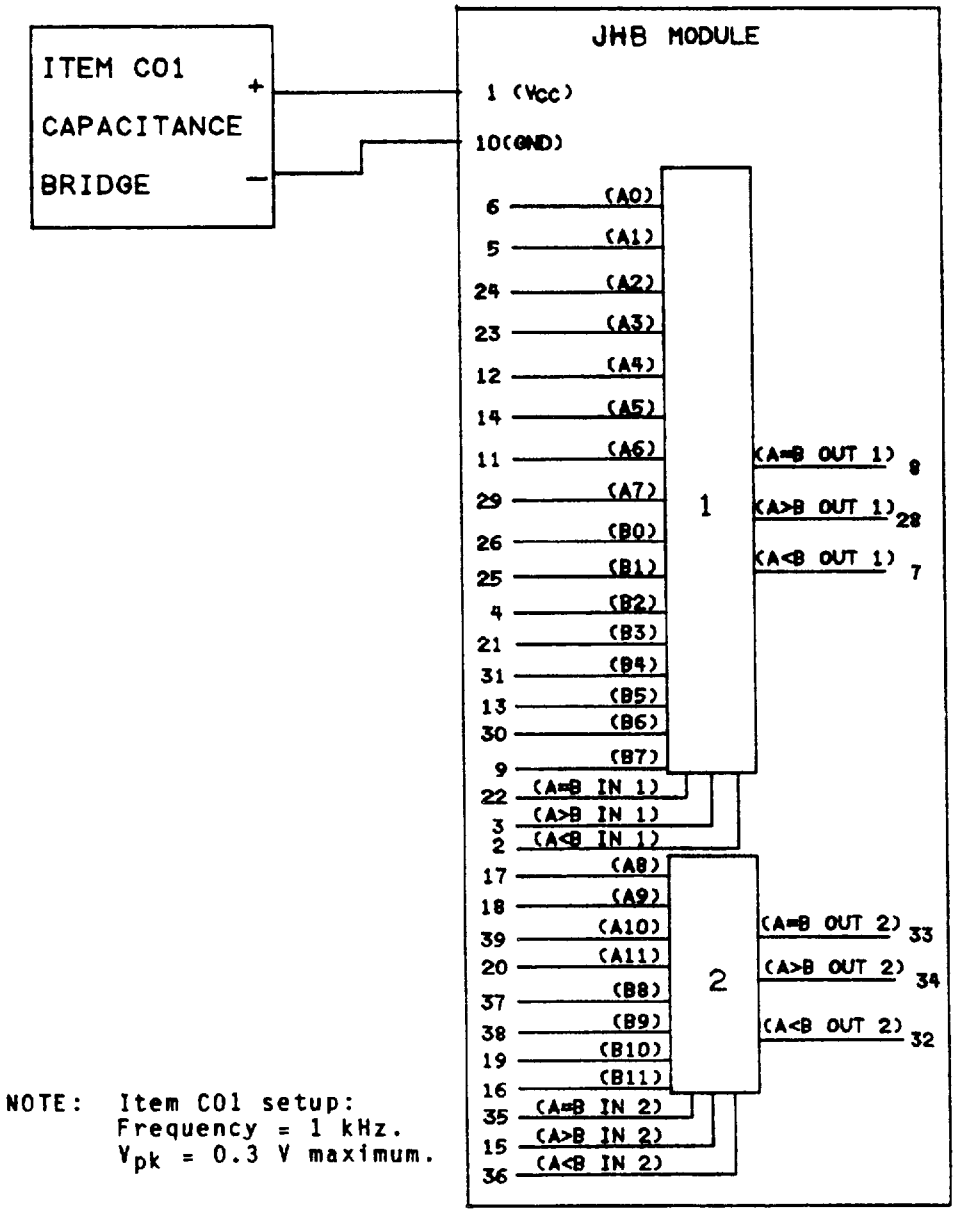
FIGURE 15. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XII.

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TABLE XIII. Power supply filter capacitance test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
P	Power supply filter capacitance	1.6	3.0	µF	16

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NOTE: Item C01 setup:
Frequency = 1 kHz.
V_{pk} = 0.3 V maximum.

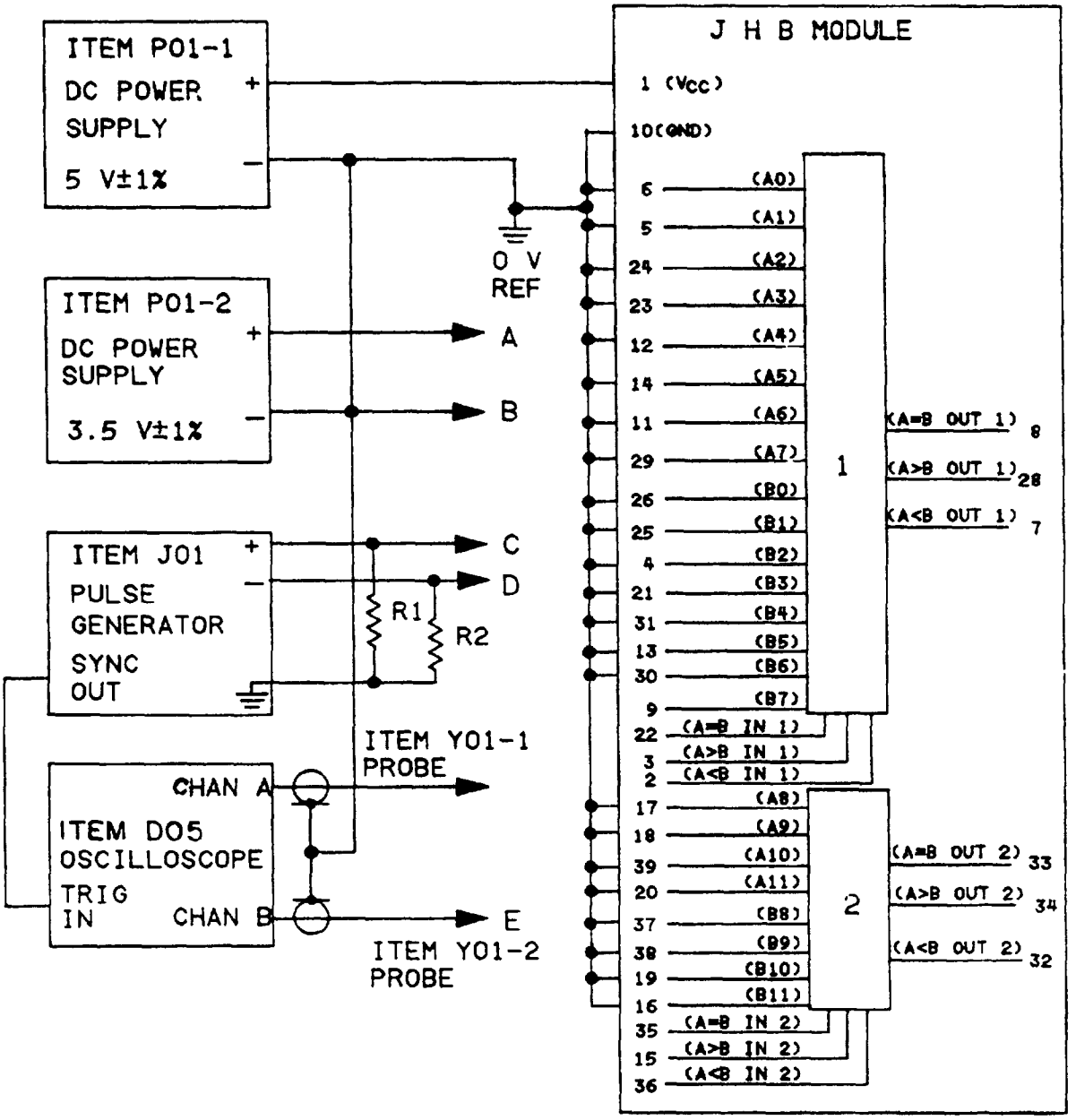
FIGURE 16. Power supply filter capacitance test circuit for table XIII.

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TABLE XIV. Output crosstalk (high-level and low-level) test.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Q1	A>B OUT 1	Output crosstalk, high level	3	2,15,22	35	36	28	0	400	mV	17,18
Q2	A<B OUT 1	"	2	3,22,36	35	15	7	"	"	"	"
Q3	A=B OUT 1	"	22	2,3,35	36	15	8	"	"	"	"
Q4	A>B OUT 2	"	15	3,35,36	22	2	34	"	"	"	"
Q5	A<B OUT 2	"	36	2,15,35	22	3	32	"	"	"	"
Q6	A=B OUT 2	"	35	15,22,36	2	3	33	"	"	"	"
R1	A>B OUT 1	Output crosstalk, low level	22	2,3,15	35	36	28	"	500	"	"
R2	A<B OUT 1	"	22	2,3,36	35	15	7	"	"	"	"
R3	A=B OUT 1	"	2	3,22,35	36	15	8	"	"	"	"
R4	A>B OUT 2	"	35	15,3,36	22	2	34	"	"	"	"
R5	A<B OUT 2	"	35	2,15,36	22	3	32	"	"	"	"
R6	A=B OUT 2	"	36	15,22,35	2	3	33	"	"	"	"

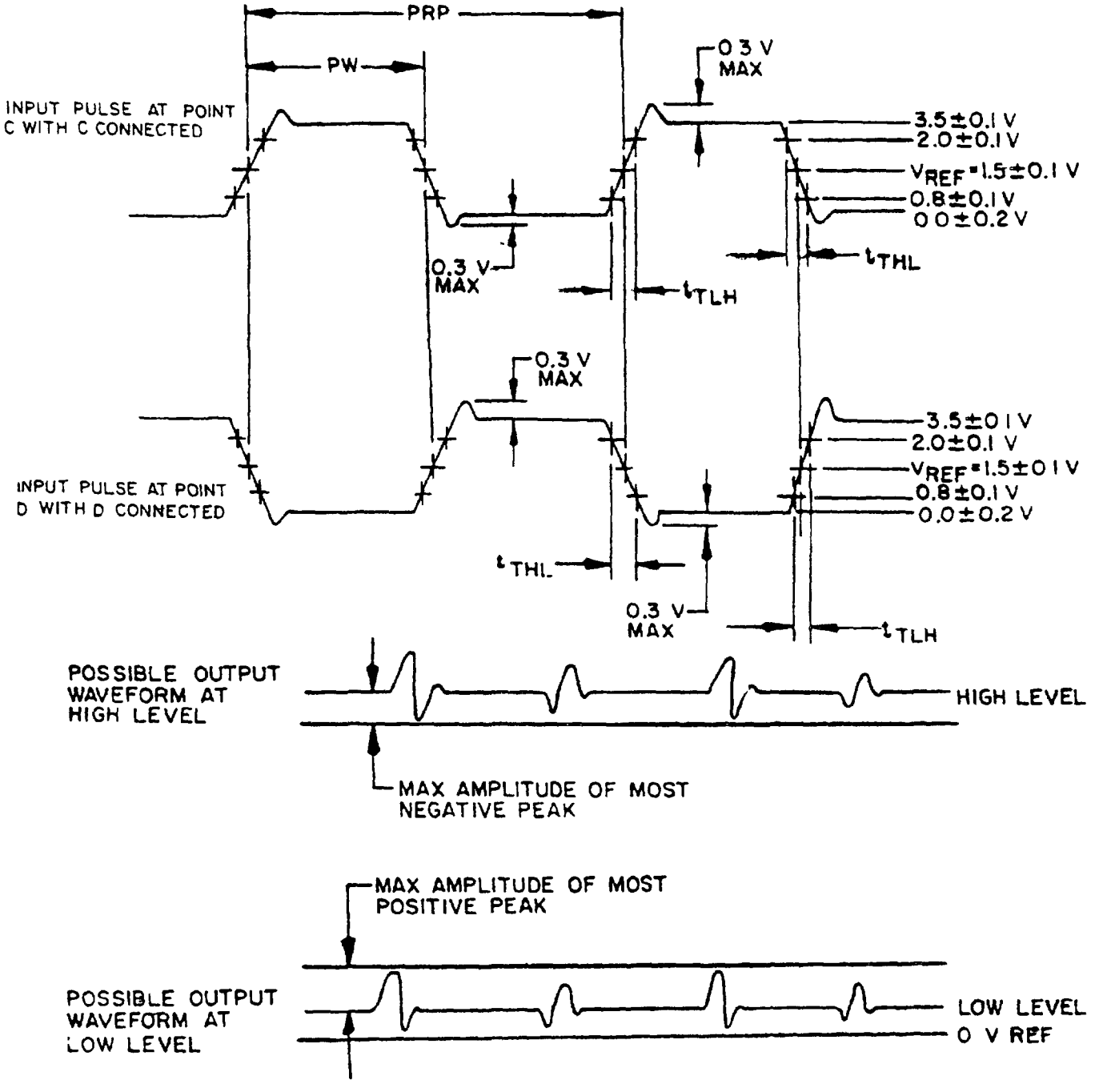
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. Load each output with a load as shown on figure 12.

FIGURE 17. Output crosstalk (high-level and low-level) test circuit for table XIV.

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NOTES:
1. Input pulse parameters:
   a. $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns (verified at each input pin).
   b. PW = 500 ±50 ns.
   c. PRP = 1,000 ±100 ns.
2. The lead length of electrical conductors between resistor R1 and input pins, items YO1-1 and YO1-2 to input and output pins, load circuits to output pins, and ground leads of items YO1-1 and YO1-2 to test fixture and module ground pins shall be less than 2 inches (50.8 mm).

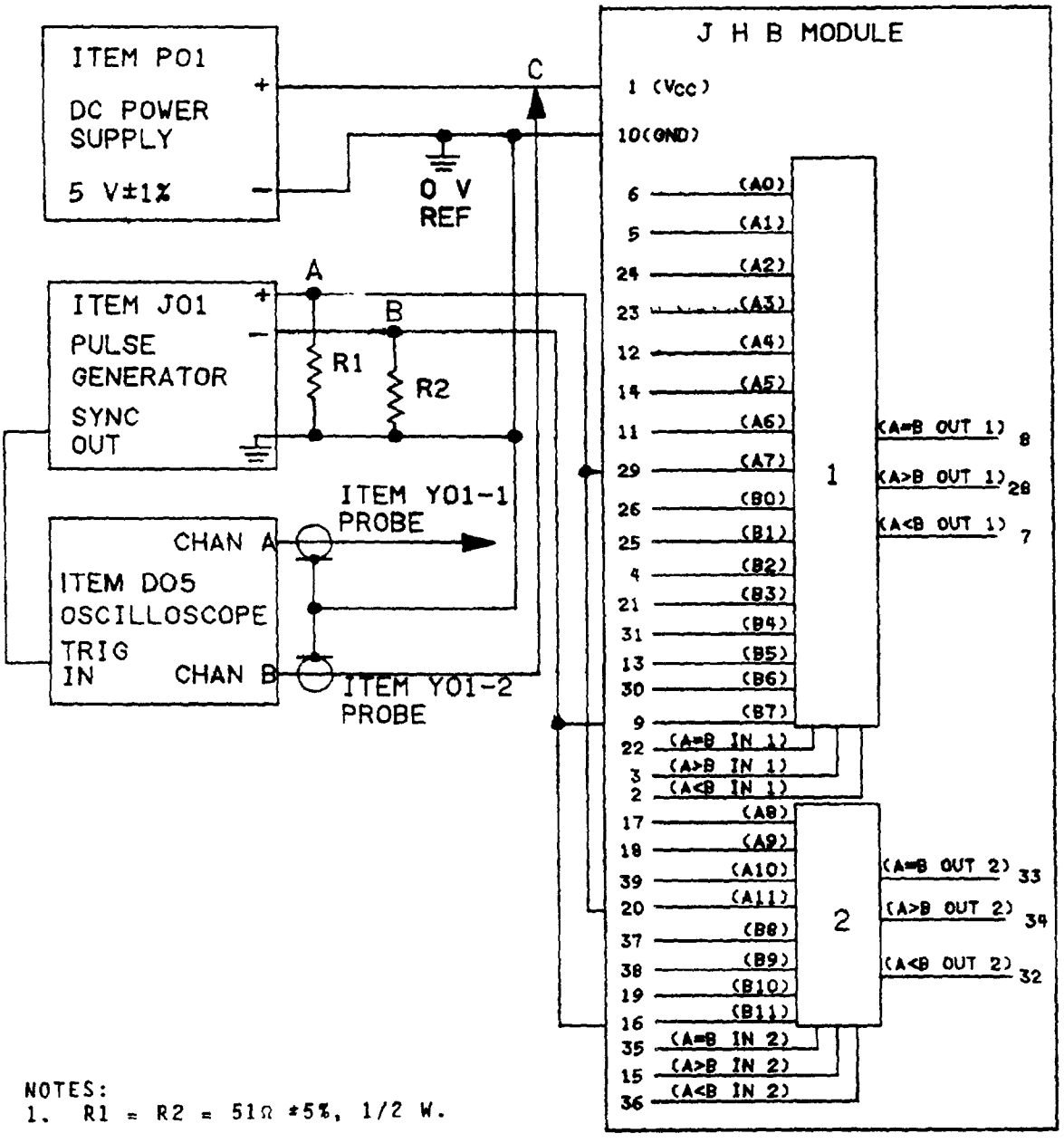
FIGURE 18. Output crosstalk (high-level and low-level) test circuit waveforms for table XIV.

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TABLE XV. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
S	Power supply transient amplitude	0	500	mV	19,20

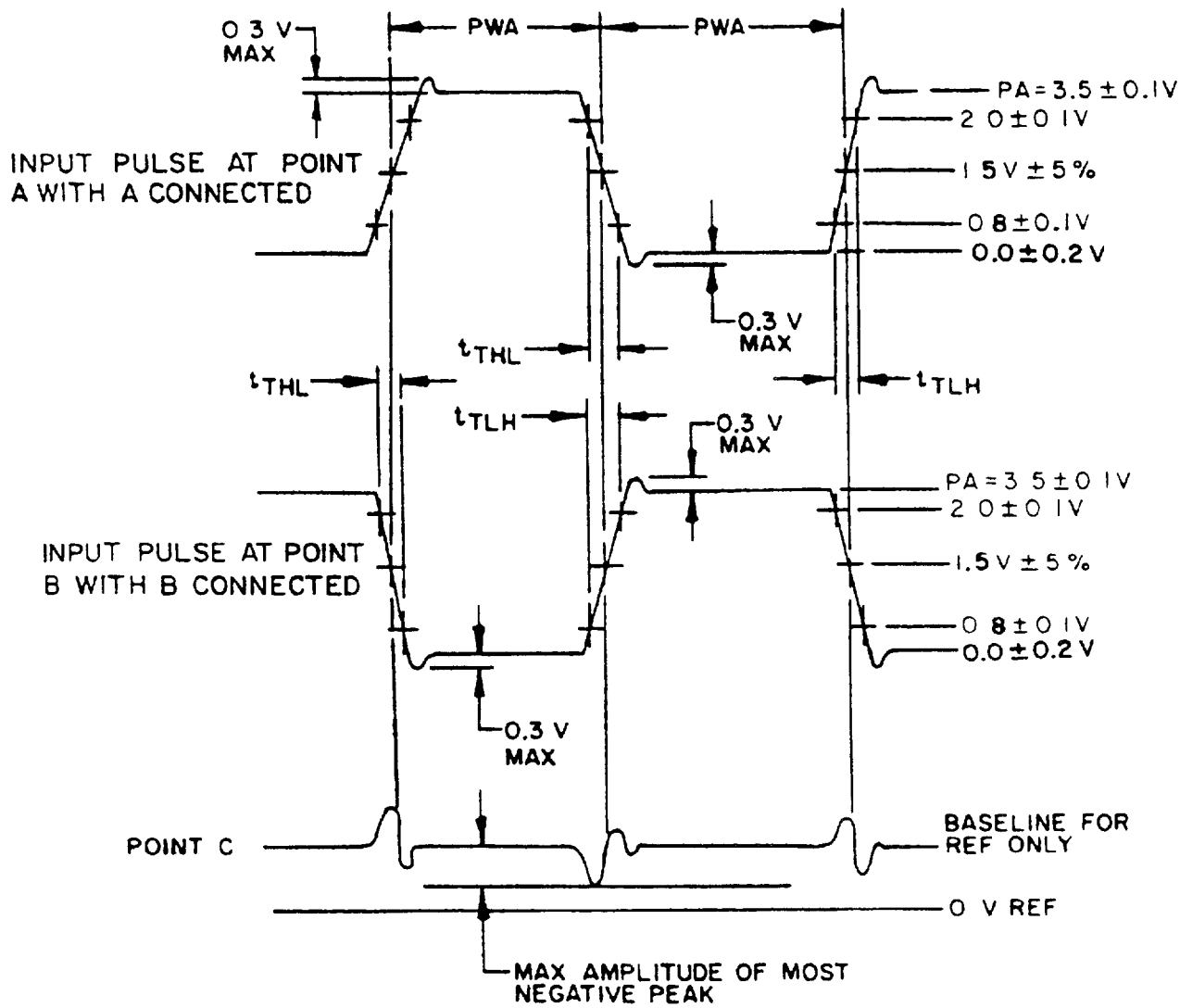
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.

FIGURE 19. Power supply transient amplitude test circuit for table XV.

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NOTE: Input pulse parameters:

PWA = 500 ±50 ns.
$t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin).

FIGURE 20. Power supply transient amplitude test circuit waveforms for table XV.

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TABLE XVI. Input capacitance test.

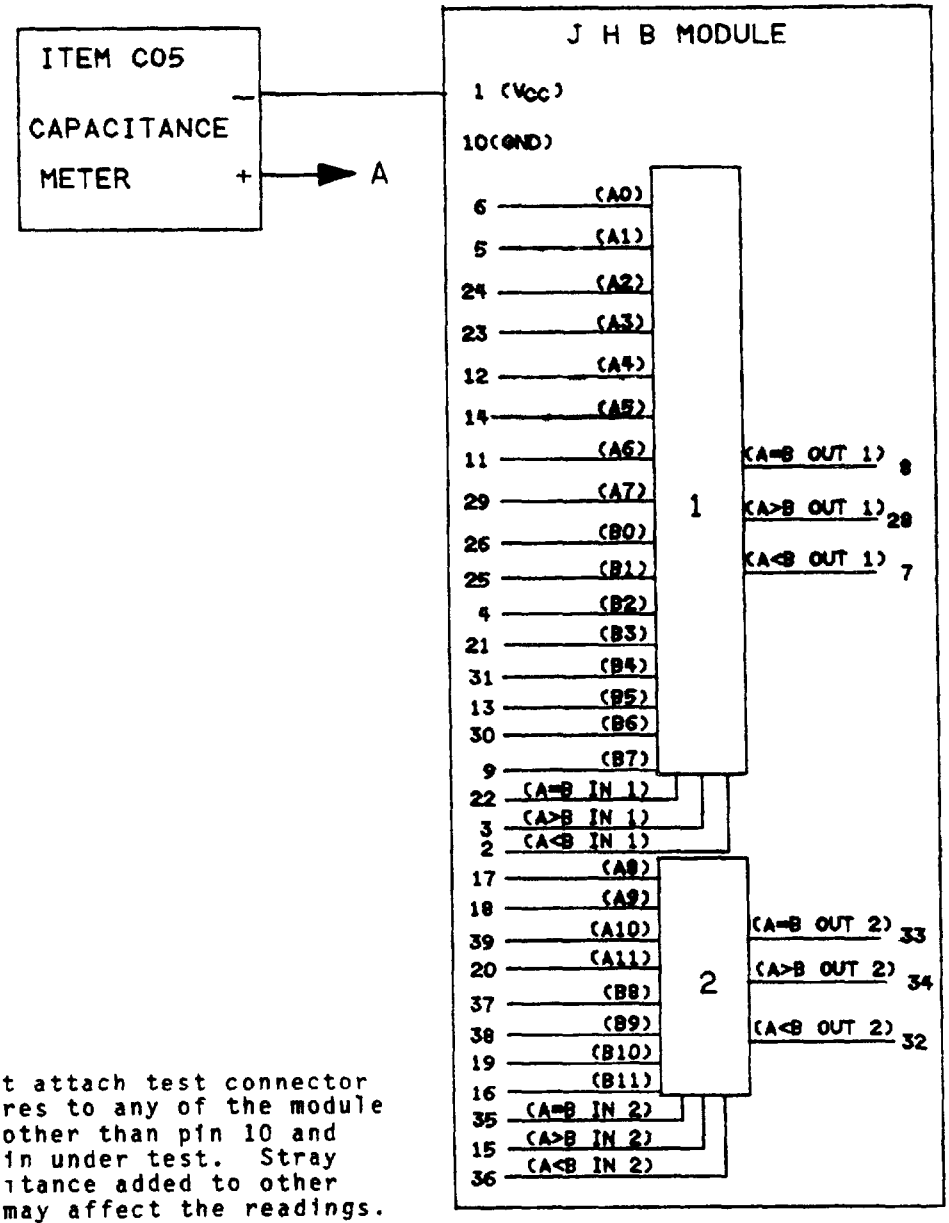
Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
T1	A>B IN 1 Input capacitance	3	2	15	pF	21
T2	A<B IN 1 "	2	"	"	"	"
T3	A>B IN 2 "	15	"	"	"	"
T4	A<B IN 2 "	36	"	"	"	"
T5	A=B IN 1 "	22	"	20	"	"
T6	A=B IN 2 "	35	"	"	"	"
T7	A0 "	6	"	"	"	"
T8	A1 "	5	"	"	"	"
T9	A2 "	24	"	"	"	"
T10	A3 "	23	"	"	"	"
T11	A4 "	12	"	"	"	"
T12	A5 "	14	"	"	"	"
T13	A6 "	11	"	"	"	"
T14	A7 "	29	"	"	"	"
T15	A8 "	17	"	"	"	"
T16	A9 "	18	"	"	"	"
T17	A10 "	39	"	"	"	"
T18	A11 "	20	"	"	"	"
T19	B0 "	26	"	"	"	"
T20	B1 "	25	"	"	"	"
T21	B2 "	4	"	"	"	"
T22	B3 "	21	"	"	"	"

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TABLE XVI. Input capacitance test - Continued.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
T23	B4 Input capacitance	31	2	20	pF	21
T24	B5 "	13	"	"	"	"
T25	B6 "	30	"	"	"	"
T26	B7 "	9	"	"	"	"
T27	B8 "	37	"	"	"	"
T28	B9 "	38	"	"	"	"
T29	B10 "	19	"	"	"	"
T30	B11 "	16	"	"	"	"

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NOTES
1. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
2. Null out the effects of the test leads prior to connecting the module.

FIGURE 21. Input capacitance test circuit for table XVI.

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